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Method and apparatus for encoding and decoding of low density parity check codes

Abstract

An encoding apparatus is provided. The encoding includes a low density parity check (LDPC) encoder which performs LDPC encoding on input bits based on a parity-check matrix to generate an LDPC codeword formed of 64,800 bits, in which the parity-check matrix includes an information word sub-matrix and a parity sub-matrix, the information word sub-matrix is formed of a group of a plurality of column blocks each including 360 columns, and the parity-check matrix and the information word sub-matrix are defined by various tables which represent positions of value one (1) present in every 360-th column.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. application Ser. No. 17/688,348 filed Mar. 7, 2022, which is a continuation of U.S. application Ser. No. 16/523,369 filed Jul. 26, 2019, which is a continuation of U.S. application Ser. No. 14/303,834 filed Jun. 13, 2014 which is now U.S. Pat. No. 10,411,736 issued Sep. 10, 2019, which claims priority from Korean Patent Application Nos. 10-2014-0058599, filed on May 15, 2014, in the Korean Intellectual Property Office, and U.S. Provisional Application Nos. 61/862,208, 61/841,502, and 61/835,096, filed on Aug. 5, 2013, Jul. 1, 2013, and Jun. 14, 2013, respectively, in the United States Patent and Trademark Office, the disclosures of which are incorporated herein by reference in their entirety.

BACKGROUND

1. Field

(1) Apparatuses and methods consistent with the exemplary embodiments of the inventive concept relate to encoding and decoding Low Density Parity Check (LDPC) codes, and more particularly, to encoding and decoding LDPC codes which perform LDPC encoding and decoding based on a parity-check matrix.

2. Description of the Related Art

(2) In a communication/broadcasting system, link performance may considerably deteriorate due to various types of noises, a fading phenomenon, and inter-symbol interference (ISI) of a channel. Therefore, to implement high-speed digital communication/broadcasting systems requiring high data throughput and reliability like next-generation mobile communications, digital broadcasting, and portable Internet, it has been required to develop technologies to overcome the noises, the fading, and the inter-symbol interference. As part of studies to overcome the noises, etc., a study on an error-correcting code which is a method for increasing reliability of communications by efficiently recovering distorted information has been actively conducted recently.

(3) LDPC codes which were first introduced by Gallager in the 1960s remain forgotten for a very long time due to complexity which could hardly be implemented at the technology level at that time.

However, as performance of turbo codes proposed by Berrou, Glavieux, and Thitimajshima in 1993 approaches Shannon's channel capacity, many studies on channel encoding based on iterative decoding and a graph thereof by performing many different interpretations on performance and characteristics of the turbo codes have been conducted. As a result, as the LDPC codes in the late 1990s are restudied, when the LDPC codes are decoded by applying sum-product algorithm based iterative decoding to the LDPC codes on a tanner graph corresponding to the LDPC codes, it was found that the performance of the LDPC codes approaches the Shannon's channel capacity.

(4) The LDPC codes may be generally defined by a parity-check matrix and represented using a bipartite graph commonly referred to as the tanner graph.

(5) Hereinafter, a systematic LDPC codeword will be described with reference to FIG. 1. The LDPC codes are LDPC encoded by receiving an information word **102** formed of $K_{\text{sub.ldpc}}$ bits or symbols to generate a codeword **100** formed of $N_{\text{sub.ldpc}}$ bits or symbols. Hereinafter, for convenience of explanation, it is assumed that the codeword **100** formed of $N_{\text{sub.ldpc}}$ bits is generated by receiving the information word **102** including $K_{\text{sub.ldpc}}$ bits. That is, when the information word $l = [i_{\text{sub.0}}, i_{\text{sub.1}}, i_{\text{sub.2}}, \dots, i_{\text{sub.K}_{\text{sub.ldpc.sub.-1}}}]$ **102** which is formed of $K_{\text{sub.ldpc}}$ input bits is LDPC encoded, the codeword $c = [c_{\text{sub.0}}, c_{\text{sub.1}}, c_{\text{sub.2}}, c_{\text{sub.3}}, \dots, c_{\text{sub.N}_{\text{sub.ldpc.sub.-1}}}]$ **100** is generated. That is, the codeword is a bit string formed of a plurality of bits and the bits of the codeword represent each bit forming the codeword. Further, the information word is a bit string formed of a plurality of bits and the bits of the information word represent each bit forming the information word. In this case, in the case of a systematic code, the codeword is formed of $c = [c_{\text{sub.0}}, c_{\text{sub.1}}, c_{\text{sub.2}}, \dots, c_{\text{sub.N}_{\text{sub.ldpc.sub.-1}}}] = [i_{\text{sub.0}}, i_{\text{sub.1}}, i_{\text{sub.2}}, \dots, i_{\text{sub.K}_{\text{sub.ldpc.sub.-1}}}, p_{\text{sub.0}}, p_{\text{sub.1}}, p_{\text{sub.2}}, \dots]$

$\dots, p_{\text{sub}.N_{\text{ldpc.sub.}}-1}$. Here, $P=[p_{\text{sub}.0}, p_{\text{sub}.1}, p_{\text{sub}.2}, \dots, p_{\text{sub}.N_{\text{ldpc.sub.}}-1}]$ is a parity **104** and the number $N_{\text{sub}.parity}$ of parity bits is as follows.
 $N_{\text{sub}.parity}=N_{\text{sub}.ldpc}-K_{\text{sub}.ldpc}$.

(6) The LDPC codes are a kind of linear block codes and include a process of determining a codeword satisfying conditions of following mathematical expression 1.

$$H_{\text{Math.c.sup.}}T=[h_{\text{sub}.0}, h_{\text{sub}.1}, h_{\text{sub}.2}, \dots, h_{\text{sub}.N_{\text{ldpc.sub.}}-1}]_{\text{Math.c.sup.}}T=\sum_{i=0}^{N_{\text{ldpc.sub.}}-1} c_{\text{sub}.i} H_{\text{sub}.i} \quad (1),$$

where $c=[c_{\text{sub}.0}, c_{\text{sub}.1}, c_{\text{sub}.2}, \dots, c_{\text{sub}.N_{\text{ldpc.sub.}}-1}]$.

(7) In mathematical expression 1 above, H represents the parity-check matrix, C represents the codeword, $c_{\text{sub}.i}$ represents an i-th bit of the codeword, and $N_{\text{sub}.ldpc}$ represents a codeword length. Here, $h_{\text{sub}.i}$ represents an i-th column of the parity-check matrix H.

(8) The parity-check matrix H is formed of the same $N_{\text{sub}.ldpc}$ columns as the number of bits of the LDPC codeword. The mathematical expression 1 represents that since a sum of a product of the i-th column $h_{\text{sub}.i}$ of the parity-check matrix and the i-th bit $c_{\text{sub}.i}$ of the codeword becomes “0”, the i-th column $h_{\text{sub}.i}$ has a relationship with the i-th bit $c_{\text{sub}.i}$ of the codeword.

(9) Meanwhile, the performance of the LDPC codes may be determined according to the parity-check matrix. Therefore, there is a need to design the parity-check matrix for the LDPC codes having improved performance.

SUMMARY

(10) One or more exemplary embodiments may overcome the above disadvantages and other disadvantages not described above. However, these embodiments are not required to overcome the disadvantages described above and may not overcome any of the problems described above.

(11) One or more exemplary embodiments provide a method and an apparatus for encoding and decoding LDPC codes capable of improving LDPC encoding and decoding performance.

(12) According to an aspect of an exemplary embodiment, there is provided an encoding apparatus which may include: an LDPC encoder configured to perform LDPC encoding on input bits based on a first parity-check matrix or a second parity-check matrix to generate an LDPC codeword formed of 64,800 bits, wherein the parity-check matrix includes an information word sub-matrix and a parity sub-matrix, and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.

(13) The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 17 when a code rate is 6/15.

(14) The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

(15) The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

(16) The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

(17) According to an aspect of another exemplary embodiment, there is provided an encoding method which may include: generating an LDPC codeword formed of 64,800 bits by performing LDPC encoding on input bits based on a parity-check matrix, in which the parity-check matrix includes an information word sub-matrix and a parity sub-matrix and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.

(18) In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 17 when a code rate is 6/15.

(19) In the generating of the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

(20) In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

(21) In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

- (22) According to an aspect of still another exemplary embodiment, there is provided a decoding apparatus which may include: an LDPC decoder configured to perform LDPC decoding on an LDPC codeword formed of 64,800 bits based on a parity-check matrix, wherein the parity-check matrix includes an information word sub-matrix and a parity sub-matrix, and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.
- (23) The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 17 when the code rate is 6/15.
- (24) The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.
- (25) The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.
- (26) The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.
- (27) According to an aspect of still another exemplary embodiment, there is provided a decoding method which may include: performing LDPC decoding on an LDPC codeword formed of 64,800 bits based on a parity-check matrix, in which the parity-check matrix includes an information word sub-matrix and a parity sub-matrix and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) present in every 360-th column.
- (28) In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 17 when the code rate is 6/15.
- (29) In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.
- (30) In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.
- (31) In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.
- (32) Additional and/or other aspects and advantages of the exemplary embodiments will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of these embodiments.
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Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) The above and/or other aspects will be more apparent by describing certain exemplary embodiments with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a diagram illustrating a codeword of a systematic LDPC code;
- (3) FIG. 2 is a diagram illustrating a parity-check matrix and a factor graph of general (8, 2, 4) LDPC codes.
- (4) FIG. 3 is a diagram illustrating a parity-check matrix according to an exemplary embodiment;
- (5) FIG. 4 is a diagram illustrating the parity-check matrix generated by permuting a row and a column of the parity-check matrix of FIG. 3, according to an exemplary embodiment;
- (6) FIGS. 5A and 5B are diagrams illustrating a check node and a variable node used for LDPC decoding, according to exemplary embodiments;
- (7) FIG. 6 is a block diagram for describing a configuration of an encoding apparatus, according to an exemplary embodiment;
- (8) FIG. 7 is a block diagram for describing a configuration of a transmitting apparatus, according to an exemplary embodiment;
- (9) FIG. 8 is a block diagram for describing a configuration of a decoding apparatus, according to an exemplary embodiment;
- (10) FIG. 9 is a block diagram for describing a decoding apparatus, according to an exemplary

embodiment;

(11) FIG. 10 is a block diagram for describing a configuration of a receiving apparatus, according to an exemplary embodiment;

(12) FIGS. 11 and 12 are diagrams for describing an effect generated at the time of performing LDPC encoding, according to exemplary embodiments;

(13) FIG. 13 is a flow chart for describing an encoding method, according to an exemplary embodiment; and

(14) FIG. 14 is a flow chart for describing a decoding method, according to an exemplary embodiment.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(15) Hereinafter, exemplary embodiments will be described in detail with reference to the accompanying drawings. Further, in describing the exemplary embodiments, detailed descriptions of well-known functions or constructions will be omitted so as not to obscure the description with unnecessary detail.

(16) Hereinafter, the exemplary embodiments will describe a technology of LDPC encoding and LDPC decoding in a communication/broadcasting system.

(17) Hereinafter, the exemplary embodiments use terms and names which are defined in the Digital Video Broadcasting the Second Generation Terrestrial (DVB-T2) system which is one of the European digital broadcasting standards and North America digital broadcasting standard system, Advanced Television Systems Committee (ATSC) 3.0 which is being established as standard. However, the inventive concept is not limited to these terms and names, but may be similarly applied to other systems.

(18) A graph representation method of LDPC codes will be described with reference to FIG. 2.

(19) FIG. 2 is a diagram illustrating an example of a parity-check matrix $H_{\text{sub}.1}$ of the LDPC codes which is formed of four (4) rows and eight (8) columns. The parity-check matrix $H_{\text{sub}.1}$ is represented with a tanner graph. Referring to FIG. 2, the parity-check matrix $H_{\text{sub}.1}$ has eight (8) columns, and thus, a codeword having a length of eight (8) is generated. The Codes generated through the $H_{\text{sub}.1}$ represent the LDPC codes, and each column of the $H_{\text{sub}.1}$ corresponds to encoded eight (8) bits.

(20) Referring to FIG. 2, the tanner graph of the LDPC codes which are encoded and decoded based on the parity-check matrix $H_{\text{sub}.1}$ is formed of eight (8) variable nodes, that is, $x_{\text{sub}.1}$ 202, $x_{\text{sub}.2}$ 204, $x_{\text{sub}.3}$ 206, $x_{\text{sub}.4}$ 208, $x_{\text{sub}.5}$ 210, $x_{\text{sub}.6}$ 212, $x_{\text{sub}.7}$ 214 and $x_{\text{sub}.8}$ 216 and four check nodes 218, 220, 222 and 224. Here, an i -th column and a j -th row of the parity-check matrix $H_{\text{sub}.1}$ of the LDPC codes each correspond to a variable node $x_{\text{sub}.i}$ and a j -th check node, respectively. Further, a value of one (1) at an intersection point of a j -th column and a j -th row of the parity-check matrix $H_{\text{sub}.1}$ of the LDPC codes, that is, a non-zero value represents that an edge connecting between the variable node $x_{\text{sub}.i}$ and the j -th check node on the tanner graph as illustrated in FIG. 2 exists.

(21) A degree of the variable node and the check node on the tanner graph of the LDPC codes represents the number of edges connected to each node, which is the same as the number of non-zero entries in a row or a column corresponding to the corresponding node in the parity-check matrix of the LDPC codes. For example, in FIG. 2, the degrees of the variable nodes $x_{\text{sub}.1}$ 202, $x_{\text{sub}.2}$ 204, $x_{\text{sub}.3}$ 206, $x_{\text{sub}.4}$ 208, $x_{\text{sub}.5}$ 210, $x_{\text{sub}.6}$ 212, $x_{\text{sub}.7}$ 214 and $x_{\text{sub}.8}$ 216 each become 4, 3, 3, 3, 2, 2, 2 and 2 in order, and the degrees of the check nodes 218, 220, 222 and 224 each become 6, 5, 5 and 5 in order. Further, the number of non-zero entries in each column of the parity-check matrix $H_{\text{sub}.1}$ of FIG. 2 corresponding to the variable nodes of FIG. 2 coincides with the degrees 4, 3, 3, 3, 2, 2, 2 and 2 in order, and in each row of the parity-check matrix $H_{\text{sub}.1}$ of FIG. 2 corresponding to the check nodes of FIG. 2, the number of non-zero entries coincides with the degrees 6, 5, 5 and 5 in order.

(22) The LDPC codes may be decoded using an iterative decoding algorithm based on a sum-product algorithm on a bipartite graph illustrated in FIG. 2. Herein, the sum-product algorithm is a kind of message passing algorithm, and the message passing algorithm represents an algorithm which exchanges messages through the edge on the bipartite graph and calculates output messages using messages input to the variable nodes or the check nodes to update the messages.

(23) Herein, a value of an i -th encoding bit may be determined based on a message of an i -th variable node. A soft decision and a hard decision may be performed on the value of the i -th encoding bit. Therefore, performance of an i -th bit $c_{\text{sub}.i}$ of the LDPC codeword corresponds to performance of the i -th variable node, which may be determined depending on positions and the number of ones (1s) in the i -th column of the parity-check matrix. That is, performance of $N_{\text{sub}.ldpc}$ bits of the codeword relies on

the positions and number of ones (1s) in the parity-check matrix.

(24) Hereinafter, characteristics of the parity-check matrix of the LDPC codes having a specific structure will be described with reference to FIG. 3.

(25) FIG. 3 illustrates a parity-check matrix having a structure according to an exemplary embodiment. The parity-check matrix illustrated in FIG. 3 has a structure which may encode and decode a systematic code in which the codeword includes an original information word. Hereinafter, according to an exemplary embodiment will be described based on the parity-check matrix of FIG. 3, but a scope of application of the inventive concept is not limited to the parity-check matrix as illustrated in FIG. 3.

(26) In FIG. 3, N.sub.ldpc represents the LDPC codeword length and K.sub.ldpc represents the information word length. Meanwhile, the codeword length or the information word length represents the number of bits included in the codeword or the information word, respectively. M represents an interval at which a pattern of a column is repeated in a sub-matrix 310 (hereinafter, referred to as an information word sub-matrix) corresponding to the information word and Q.sub.ldpc is a size of cyclic shift of each column in the information word sub-matrix 310, in which values of the integer M and the Q.sub.ldpc are determined to be $Q_{\text{sub_ldpc}} = (N_{\text{sub_ldpc}} - K_{\text{sub_ldpc}}) / M$. Here, $K_{\text{sub_ldpc}} / M$ is also an integer. Values of the M and the Q.sub.ldpc may be changed depending on the codeword length and a code rate (or coding rate).

(27) Referring to FIG. 3, the parity-check matrix 300 is divided into the information word sub-matrix 310 (or information word matrix) corresponding to the information word and a parity sub-matrix 320 (or parity matrix) corresponding to the parity. The information word sub-matrix 310 includes K.sub.ldpc columns and the parity sub-matrix 320 includes $N_{\text{sub_parity}} = N_{\text{sub_ldpc}} - K_{\text{sub_ldpc}}$ columns. The number of rows of the parity-check matrix 300 and the number of columns of the parity sub-matrix 320 are equal to $N_{\text{sub_ldpc}} - K_{\text{sub_ldpc}}$.

(28) Positions of entries having weight-1s, that is, value one (1) present in the parity sub-matrix 320 including a K.sub.ldpc-th column to an $(N_{\text{sub_ldpc}} - 1)$ -th column of the parity-check matrix 300 have a dual diagonal structure. Therefore, among the columns included in the parity sub-matrix 320, all the degrees (herein, the degree is the number of ones (1s) included in each column) of the remaining columns except the $(N_{\text{sub_ldpc}} - 1)$ -th column are two (2) and the degree of the $(N_{\text{sub_ldpc}} - 1)$ -th column is one (1).

(29) Meanwhile, a structure of the information word sub-matrix 310, that is, the sub-matrix including a zero (0)-th column to a $(K_{\text{sub_ldpc}} - 1)$ -th column depends on the following rule.

(30) First, the K.sub.ldpc columns corresponding to the information word in the parity-check matrix 300 belong to a plurality of column groups each having M columns, and the information word sub-matrix 310 is divided into a total of $K_{\text{sub_ldpc}} / M$ column groups. In columns belonging to the same column group, positions of rows at which one (1) is present have a relationship in which a row having a value of one (1) is shifted by Q.sub.ldpc from an immediately previous row having the same value of one (1). Here, Q.sub.ldpc may be an integer greater than or equal to one (1).

(31) Second, if it is assumed that a degree of a zero (0)-th column of an i-th ($i=0, 1, \dots, K_{\text{sub_ldpc}} / M - 1$) column group is D.sub.i and positions of each row at which one (1) is positioned are $R_{\text{sub_i},0,\text{sup.}}(0), R_{\text{sub_i},0,\text{sup.}}(1), \dots, R_{\text{sub_i},0,\text{sup.}}(D_{\text{sub_i}} - 1)$, an index $R_{\text{sub_i},j,\text{sup.}}(k)$ of a row at which a k-th weight-1, i.e., k-th one (1), is positioned in a j-th column of the i-th column group is determined as represented by following mathematical expression 2.

$$R_{\text{sub_i},j,\text{sup.}}(k) = (R_{\text{sub_i},(i-1),\text{sup.}}(k) + Q_{\text{sub_ldpc}}) \bmod (N_{\text{sub_ldpc}} - K_{\text{sub_ldpc}}) \quad (2)$$

(32) In mathematical expression 2 above, $k=0, 1, 2, \dots, D_{\text{sub_i}} - 1$, $i=0, 1, \dots, K_{\text{sub_ldpc}} / M - 1$, and $j=1, 2, \dots, M - 1$.

(33) The above mathematical expression 2 may be represented like following mathematical expression 3.

$$R_{\text{sub_i},j,\text{sup.}}(k) = (R_{\text{sub_i},0,\text{sup.}}(k) + (j \bmod M) \times Q_{\text{sub_ldpc}}) \bmod (N_{\text{sub_ldpc}} - K_{\text{sub_ldpc}}) \quad (3)$$

(34) In mathematical expression 3 above, $k=0, 1, 2, \dots, D_{\text{sub_i}} - 1$, $i=0, 1, \dots, K_{\text{sub_ldpc}} / M - 1$, and $j=1, 2, \dots, M - 1$.

(35) In the above mathematical expressions, $R_{\text{sub_i},j,\text{sup.}}(k)$ represent the indices of the row at which the k-th weight-1 is positioned in the j-th column of the i-th column group, N.sub.ldpc represents the LDPC codeword length, K.sub.ldpc represents the information word length, D.sub.i represents the degree of a zero (0)-th column belonging to the i-th column group, and M represents the number of columns

belonging to one column group.

(36) According to the above mathematical expressions, when only a value of $R_{sub.i,0.sup.}(k)$ is known, the indices of the row at which the k -th weight-1 is positioned in the i -th column group may be known. Therefore, when index values of the row at which the k -th weight-1 is positioned in the zero (0)-th column of each column group are stored, the positions of the column and the row at which the weight-1 is positioned may be understood in the parity-check matrix **300** (that is, the information word sub-matrix **310** of the parity-check matrix **300**) having the structure illustrated in FIG. 3.

(37) According to the above-mentioned rules, all of the degrees of the columns belonging to the i -th column group are D_i . According to the above-mentioned rules, the LDPC codes in which information on the parity-check matrix **300** is stored may be simply represented as follows.

(38) As a specific example, when $N_{sub.ldpc}$ is 30, $K_{sub.ldpc}$ is 15, and $Q_{sub.ldpc}$ is three (3), position information of rows at which the weight-1 is positioned in zero (0)-th columns of three column groups may be represented by sequences as represented by following mathematical expression 4. The sequences represented by mathematical expression 4 may be called a weight-1 position sequence.

$R_{sub.1,0.sup.}(1)=1, R_{sub.1,0.sup.}(2)=2, R_{sub.1,0.sup.}(3)=8, R_{sub.1,0.sup.}(4)=10,$

$R_{sub.2,0.sup.}(1)=0, R_{sub.2,0.sup.}(2)=9, R_{sub.2,0.sup.}(3)=13,$

$R_{sub.3,0.sup.}(1)=0, R_{sub.3,0.sup.}(2)=14. \quad (4)$

(39) In mathematical expression 4 above, $R_{sub.i,j.sup.}(k)$ represents the indices of the row at which the k -th weight-1 is positioned in the j -th column of the i -th column group.

(40) The weight-1 position sequences as represented by the above mathematical expression 4, which represents the indices of the rows at which one (1) is positioned in the zero (0)-th column of each column group, may be more simply represented like following Table 1.

(41) TABLE-US-00001 TABLE 1 1 2 8 10 0 9 13 0 14

(42) Table 1 above shows the positions of entries having the weight-1, that is, value one (1) in the parity-check matrix, in which an i -th weight-1 position sequence is represented by the indices of the row at which the weight-1 is positioned in the zero (0)-th column belonging to the i -th column group.

(43) When the column and the row of the parity-check matrix **300** illustrated in FIG. 3 are permuted by following mathematical expression 5 (row permutation) and mathematical expression 6 (column permutation), the parity-check matrix **300** illustrated in FIG. 3 may be shown as a form of a parity-check matrix **400** illustrated in FIG. 4.

$Q_{sub.ldpc} \cdot Math.i + j \cdot Math.M \cdot Math.j + i \quad (0 \leq i < M, 0 \leq j < Q_{sub.ldpc}) \quad (5)$

$i \cdot Math.i \quad (0 \leq i < K_{sub.ldpc})$

$K_{sub.ldpc} + Q_{sub.ldpc} \cdot Math.k + l \cdot Math.K_{sub.ldpc} + M \cdot Math.l + k \quad (0 \leq k < M, 0 \leq l < Q_{sub.ldpc}) \quad (6)$

(44) In the above, the row permutation represents that an order of the rows of the parity-check matrix **300** is changed using mathematical expression 5 above. Further, the column permutation represents that an order of columns of the parity-check matrix **300** is changed using mathematical expression 6 above.

(45) A method of performing permutation based on mathematical expressions 5 and 6 above is as follows. In this case, the column permutation is applied with the same principle as the row permutation, except the fact that the column permutation is applied only to the parity sub-matrix **320** by $i \cdot Math.i$ ($0 \leq i < K_{sub.ldpc}$). Hereinafter, the row permutation will be described as an example.

(46) In the case of the row permutation, i and j meeting $X = Q_{sub.ldpc} \times i + j$ for an X -th row are calculated, and the calculated i and j are substituted in $M \times j + i$ to calculate a row in which the X -th row is permuted. For example, in the case of a seventh row, i and j meeting $7 = 2 \times i + j$ each are 3 and 1 and therefore the seventh row is permuted to a thirteenth ($10 \times 1 + 3 = 13$) row.

(47) When the row and column of the parity-check matrix of FIG. 3 are each permuted depending on the mathematical expressions 5 and 6, the parity-check matrix may be shown in a form of the parity-check matrix illustrated in FIG. 4.

(48) Referring to FIG. 4, the parity-check matrix **400** of the LDPC codes has a form in which the entire parity-check matrix **400** is divided into a plurality of partial blocks and each of the partial blocks corresponds to an $M \times M$ quasi-cyclic matrix.

(49) The parity-check matrix **400** illustrated in FIG. 4 is formed in a unit of the $M \times M$ quasi-cyclic matrix, and thus, M columns may be called a column-block (or column group) and M rows may be called a row-block (or row group). That is, the parity-check matrix **400** having the form of FIG. 4 which is used

in the present exemplary embodiment is formed of $N_{\text{sub}} \cdot q_{\text{c_column}} = N_{\text{sub}} \cdot \text{ldpc} / M$ column-blocks and $N_{\text{sub}} \cdot q_{\text{c_row}} = N_{\text{sub}} \cdot \text{parity} / M$ row-blocks.

(50) Hereinafter, $M \times M$ matrices forming the parity-check matrix **400** having the form of FIG. 4 will be described in detail.

(51) First, an $M \times M$ matrix **440** of an $(N_{\text{sub}} \cdot q_{\text{c_column}} - 1)$ -th column-block among the zero (0)-th row-blocks has a form as represented by following mathematical expression 7.

$$(52) \quad A = \begin{bmatrix} 0 & 0 & \text{.Math.} & 0 & 0 \\ 1 & 0 & \text{.Math.} & 0 & 0 \\ 0 & 1 & \text{.Math.} & 0 & 0 \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ 0 & 0 & \text{.Math.} & 1 & 0 \end{bmatrix} \quad (7)$$

(53) In the $M \times M$ matrix as represented by mathematical expression 7 above, all the values of the zero (0)-th row and an $M-1$ -th column are '0'. For $0 \leq i \leq (M-2)$, an $(i+1)$ -th row of the i -th column is '1' and all the other values are '0'.

(54) Second, in the parity sub-matrix **420** of the parity-check matrix **400** of FIG. 4, for $0 \leq i \leq (N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}) / M - 1$, an i -th row-block of an $(K_{\text{sub}} \cdot \text{ldpc} / M + i)$ -th column-blocks is formed of unit matrices $I_{\text{sub}} \cdot M \times M$ **460**. Further, for $0 \leq i \leq (N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}) / M - 2$, an $(i+1)$ -th row block of the $K_{\text{sub}} \cdot \text{ldpc} / M + i$ -th column-blocks is formed of the unit matrices $I_{\text{sub}} \cdot M \times M$ **460**.

(55) Third, the information word sub-matrix **410** may be a form in which a matrix $P_{\text{sup.a.sup.ij}}$ obtained by cyclically shifting a quasi-cyclic matrix P , or the quasi-cyclic matrix P and a matrix $P_{\text{sup.a.sup.ij}}$ obtained by cyclically shifting the quasi-cyclic matrix P are combined.

(56) The quasi-cyclic matrix P is represented by following mathematical expression 8.

$$(57) \quad P = \begin{bmatrix} 0 & 1 & 0 & & 0 \\ 0 & 0 & 1 & \text{.Math.} & 0 \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & & \text{.Math.} \\ 0 & 0 & 0 & \text{.Math.} & 1 \\ 1 & 0 & 0 & & 0 \end{bmatrix} \quad (8)$$

(58) The quasi-cyclic matrix P of the above mathematical expression 8 is a square matrix having an $M \times M$ size and is a matrix of which the respective degrees of M rows and columns forming the quasi-cyclic matrix P are one (1).

(59) If a subscript $a_{\text{sub.ij}}$ of the quasi-cyclic matrix P is 0, the quasi-cyclic matrix $P_{\text{sup.0}}$ represents a unit matrix $I_{\text{sub}} \cdot M \times M$, and if the subscript $a_{\text{sub.ij}}$ of the quasi-cyclic matrix P is ∞ , the quasi-cyclic matrix $P_{\text{sup.}\infty}$ represents a zero matrix. As illustrated in FIG. 4, in the entire parity-check matrix of the LDPC codes, the total number of columns is $N_{\text{sub}} \cdot \text{ldpc} = M \times N_{\text{sub}} \cdot q_{\text{c_column}}$ and the total number of rows is $N_{\text{sub}} \cdot \text{parity} = M \times N_{\text{sub}} \cdot q_{\text{c_row}}$. That is, the parity-check matrix **400** as illustrated in FIG. 4 is formed of $N_{\text{sub}} \cdot q_{\text{c_column}}$ column-blocks and $N_{\text{sub}} \cdot q_{\text{c_row}}$ row-blocks.

(60) The parity-check matrix **400** as illustrated in FIG. 4 is represented by positions of a non-zero quasi-cycle matrix and index values at the corresponding positions.

(61) In the case of the LDPC codes, the codeword performance is determined according to the parity-check matrix. Specifically, the codeword performance may be determined according to a weight distribution and a cycle distribution of columns and rows.

(62) The weight distribution of columns represents how many columns have one (1) and how many ones (1s) are positioned in the columns, in the $N_{\text{sub}} \cdot \text{ldpc}$ columns. Further, the weight distribution of rows represents how many rows have one (1) and how many ones (1s) are positioned in the rows, in the $N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}$ rows. Further, a weight or a degree of one (1) represents the number of ones (1s) of each column and row.

(63) Herein, the weight distribution of columns and rows may be determined based on a method called density evolution (reference: Richardson, T., and URBANKE, R.: 'The capacity of low-density parity-check codes under message-passing decoding', IEEE Trans. Inf. Theory, 2001, 47, (2), pp. 599-618).

(64) In detail, in the case of using the density evolution method, when the LDPC encoding/decoding are performed based on the parity-check matrix having the given degree distribution, it may be estimated how many times of iteration is required for coding error probability to be '0' at any signal to noise ratio

(SNR). In the case of the density evolution, since it is estimated whether the coding error probability is '0' under the assumption that the codeword length is infinite, the degree distribution of the parity-check matrix may not be determined only by the density evolution if the parity-check matrix for codes having a finite length is designed.

(65) Further, the number of ones (1s) depending on the degree distribution of the parity-check matrix affects encoding/decoding complexity, and thus, the parity-check matrix needs to be designed based on the code performance which is verified not only based on a theoretical approach called the density evolution but also on the actual encoding/decoding complexity and the designed parity-check matrix.

(66) Hereinafter, a method for designing a parity-check matrix will be described in detail.

(67) The form of a parity-check matrix may be very variously present, but the inventive concept intends to design a parity-check matrix having a specific form illustrated in FIG. 3. Further, a parity-check matrix determines the degree distribution, in which a coding gain may be maximal, based on the density evolution, and by a cycle removing method, an error floor occurs in an area in which BER/FER is low. As described above, the parity-check matrix **300** having the form of FIG. 3 turns into the parity-check matrix **400** having the form of FIG. 4 by permuting the columns and the rows. In the case of the form of FIG. 4, the parity-check matrix may be designed based on the $M \times M$ matrix, and thus, may be easily designed. Therefore, according to an exemplary embodiment, a parity-check matrix is designed in the form of the parity-check matrix **400** having the form of FIG. 4, and then, the parity-check matrix **300** having the form of FIG. 3 is designed by permuting the rows and the columns.

(68) Hereinafter, the method for designing a parity-check matrix according to an exemplary embodiment will be described in detail. Step 1) The sizes $N_{\text{sub.ldpc}}$ and $K_{\text{sub.ldpc}}$ and the values of M and $Q_{\text{sub.ldpc}}$ of the parity-check matrix to be designed are determined.

(69) According to an exemplary embodiment, the foregoing parameters of the parity-check matrix may be determined as following Table 2.

(70) TABLE-US-00002 TABLE 2 code rate $N_{\text{sub.ldpc}}$ $K_{\text{sub.ldpc}}$ $N_{\text{sub.parity}}$ M $Q_{\text{sub.ldpc}}$
 $N_{\text{sub.qc_column}}$ $N_{\text{sub.qc_row}}$ 6/15 64800 25920 38880 360 108 180 108 8/15 64800 34560 30240
360 84 180 84 10/15 64800 43200 21600 360 60 180 60 12/15 64800 51840 12960 360 36 180 36

(71) In the case of the parity-check matrix **400** having the form of FIG. 4, a parity sub-matrix **420** is fixed, and thus, only the positions and distributions of one (1) in the information word sub-matrix **410** need to be determined. Further, the positions and distributions of one (1) in the information word sub-matrix **410** are formed in a unit of $M \times M$ quasi-cyclic matrix, and thus, the number and positions of quasi-cyclic matrix, which is not a zero matrix, and cyclic shift values which are the index values of the quasi-cyclic matrices are determined.

(72) Hereinafter, in step 2, the number of quasi-cyclic matrix, not the zero matrix, is determined. Step 2) The degree distribution of the parity-check matrix is determined by the density evolution method.

(73) As described above, the distribution of one (1) in the parity-check matrix dominates the performance of the LDPC codes. Therefore, according to an exemplary embodiment, the distribution of one (1) in the parity-check matrix is determined by the density evolution method. That is, a degree distribution having the highest probability for error probability to converge to a predetermined value by performing iterative decoding as many as a predetermined times, that is, a degree distribution having the lowest SNR is selected from all the possible degree distributions.

(74) In this case, restrictions are as follows. 1) The number of ones (1s) in an LDPC code affects the encoding and decoding complexity, and thus, the number of ones (1s) needs to be limited. 2) If the number of ones (1s) present in a column is various, the decoding complexity may be increased, and thus, a kind of the number of ones (1s) needs to be limited.

(75) First, all the possible degree distributions (lists) are determined based on the parity-check matrix having parameters of the above Table 2 in consideration of the foregoing restrictions. Next, after a target SNR for each code rate is determined, when the LDPC codes encoded/decoded based on the parity-check matrix having the degree distributions present in the list at the target SNR by the density evolution are decoded, it is determined how many time of iteration is required for the error probability to converge to a predetermined value.

(76) In this case, for all the degree distributions present in the list, when the error probability is converged to a predetermined value or less within the number of iteration times smaller than a specific

value, the target SNR value is adjusted to be small and the density evolution for the degree distribution is performed again.

(77) However, when the result value of the density evolution for all the degree distribution is not converged even within more iterations than the specific value, the SNR value is adjusted to be large and the density evolution for the degree distribution is performed again.

(78) Among all the degree distributions present in the list, the degree distribution in which the error probability is converged to the predetermined value or less within the low SNR and the small iterative decoding (that is, iteration number of times) is determined as the result of step 1. Step 3) The position of the non-zero quasi-cyclic matrix is determined based on the degree distribution determined in step 2.

(79) In this step, the position of the non-zero quasi-cyclic matrix is determined based on a well known PEG algorithm (X.-Y. Hu, E. Eleftheriou, and D.-M. Arnold, "Regular and irregular progressive edge-growth tanner graphs", IEEE Trans. Inf. Theory, vol. 51, no. 1, pp. 386-398, January 2005.) In this case, additional restrictions suggested by the present invention are to make the number of 1s in each row maximally uniform. Step 4) The indices of the quasi-cycle matrices need to be adjusted so as to prevent the error floor from occurring.

(80) In this step, the parity sub-matrix is fixed, and thus, is not considered. In the information word sub-matrix, only the index values of non-zero quasi-cycle matrices positioned in a predetermined row need to be changed from a column having the lowest column degree.

(81) In this step, all index values of the non-zero quasi-cycle matrices, except the parity sub-matrix 420, positioned in the same row need to have different values.

(82) The order of the rows at which the index values are changed may be various. A change in the index values is repeated until a cycle value of a minimum cycle is no more increased or the number of variable nodes having the minimum cycle is no more improved. Here, step 4 is called lifting. Step 5) The parity-check matrix designed based on FIG. 4 needs to be modified into the form of FIG. 3 by the row permutation and the column permutation.

(83) In this case, the row permutation may be performed based on following mathematical expression 9 and the column permutation may be performed based on following mathematical expression 10.

$$M.Math.i+j.Math.Q.sub.ldpc.Math.j+i \quad (0 \leq i < Q.sub.ldpc, 0 \leq j < M) \quad (9)$$

$$i.Math.i \quad (0 \leq i < K.sub.ldpc)$$

$$K.sub.ldpc+M.Math.k+l.Math.K.sub.ldpc+Q.sub.ldpc.Math.l+k \quad (0 \leq k < Q.sub.ldpc, 0 \leq l < M) \quad (10)$$

(84) Hereinafter, a process of designing a parity-check matrix will be described in more detail with reference to a case of a code rate 10/15 (=2/3) as an example. Step 1) Parameters N.sub.ldpc and K.sub.ldpc related to the size and the values of M and Q.sub.ldpc of the parity-check matrix to be designed are determined as following Table 3.

(85) TABLE-US-00003 TABLE 3 code rate N.sub.ldpc K.sub.ldpc N.sub.parity M Q.sub.ldpc
N.sub.qc_column N.sub.qc_row 10/15 64800 43200 21600 360 60 180 60 Step 2) The degree distribution of the parity-check matrix is determined based on the density evolution method.

(86) The distributions in which the error probability is highly likely to be a predetermined value at the lowest SNR as a result of the density evolution are selected from all the possible degree distributions. For example, for the selected distributions, when SNR=2.6 dB, the iteration number of times to make BER=10.sup.-5 is obtained by the density evolution method. The selected distributions and the iteration number of times obtained based on the density evolution for the selected distributions are as shown in following Table 4. In following Table 4, N (x.sub.i) represents the number of column groups or column blocks having a degree of x.sub.i.

(87) TABLE-US-00004 TABLE 4 The number of iterations Degree Degree Degree Degree Degree meeting BER = case (x1) N(x1) (x2) N(x2) (x3) N(x3) (x4) N(x4) 10.sup.-5 1 16 23 4 2 3 95 2 60 65 2 16 23 4 13 3 84 2 60 65 3 17 20 4 31 3 69 2 60 73 4 17 19 4 42 3 59 2 60 94 5 15 26 4 0 3 94 2 60 63 6 16 24 4 0 3 96 2 60 63 Step 3) The positions of the non-zero quasi-cyclic matrix is determined based on the degree distribution determined in step 2.

(88) According to an exemplary embodiment, a degree distribution of case 6 of above Table 4 is selected. Further, to meet the selected degree distribution, positions of the non-zero quasi-cycle matrix is determined.

(89) The reason of determining case 6 is that, as described above, the number of ones (1s) depending on

the degree distribution of the parity-check matrix affects encoding/decoding complexity, and thus, the parity-check matrix needs to be designed based on the code performance which is not only based on a theoretical approach called the density evolution but also on the actual encoding/decoding complexity and the designed parity-check matrix. Step 4) The indices of the quasi-cycle matrices need to be adjusted so as to prevent the error floor from occurring.

(90) Following Table 5 shows positions and index values of the quasi-cycle matrices of the parity-check matrix **400** having the form of FIG. 4 which is determined by the foregoing method. In this process, the method called the lifting (see foregoing step 4) considering the cycle characteristics is used.

(91) TABLE-US-00005 TABLE 5 Indices of column groups in which non- zero quasi-cycle matrix of i-th row is Indices of non-zero quasi-cycle matrix positioned of i-th row

0	3	6	12	14	15	23	34	61	81	87	108																						
120	179	350	122	72	90	310	344	85	245	10	12	93	0	359	1	1	49	10	13	19	38	49	79	83	106	120	121	313					
156	2	258	227	245	5	117	280	96	211	0	0	2	3	5	15	20	22	23	44	62	78	91	99	121	122	219	33	282	344	118			
250	5	238	352	230	120	0	0	3	0	8	12	16	21	23	28	56	93	98	117	122	123	211	302	339	64	228	247	147	13				
38	143	247	0	0	4	1	7	8	9	16	19	20	46	55	64	101	123	124	104	289	301	304	55	48	170	244	39	185	268	0	0		
5	2	3	12	13	18	20	24	54	80	112	114	124	125	5	355	305	99	306	88	96	288	22	52	124	0	0	6	1	6	8	14	16	17
34	40	74	102	107	125	126	309	348	241	98	41	83	3	143	140	102	227	0	0	7	4	8	10	12	19	20	35	37	48	75			
103	126	127	108	341	333	17	211	187	104	239	140	284	137	0	0	8	0	4	14	17	21	23	33	65	78	82	98	127	128				
38	165	219	233	270	282	63	281	264	225	209	0	0	9	5	6	13	18	19	21	22	41	67	76	109	128	129	8	207	200				
148	234	33	307	100	203	162	94	0	0	10	1	2	3	5	11	13	14	29	60	91	103	112	129	130	271	291	274	212	110				
57	80	180	134	208	244	337	0	0	11	6	7	8	11	15	21	23	38	58	72	74	130	131	253	35	207	278	180	122	120				
75	147	0	174	0	0	12	4	11	19	20	21	22	25	47	80	111	115	131	132	124	95	102	48	172	177	244	144	83	207				
136	0	0	13	3	5	7	14	15	17	18	37	69	83	84	110	132	133	66	211	233	54	315	342	338	208	229	357	199	143				
0	0	14	1	4	7	10	14	15	17	20	28	52	77	88	133	134	43	282	309	252	263	115	220	85	139	214	114	302	0	0	15		
0	4	5	6	8	18	39	43	71	79	104	134	135	283	20	166	310	147	229	316	359	268	15	194	0	0	16	1	4	5	18	21	22	
40	60	84	97	118	135	136	151	213	230	83	304	132	47	336	276	233	167	0	0	17	0	6	9	14	21	22	23	27	59	75			
110	136	137	70	23	169	174	230	15	19	119	25	57	40	0	0	18	1	4	11	18	21	23	31	66	73	77	91	137	138	78			
222	317	343	322	29	293	15	97	268	17	0	0	19	3	8	12	17	18	22	47	65	90	102	117	138	139	1	173	0	282	148			
282	299	122	42	212	311	0	0	20	0	2	11	13	15	22	24	42	49	75	88	139	140	8	186	278	1	309	186	57	294	255			
293	67	0	0	21	0	5	7	9	12	16	17	45	51	84	108	117	140	141	141	342	45	270	83	272	333	19	105	153	314				
134	0	0	22	2	5	10	11	15	17	33	43	59	72	87	141	142	280	99	183	227	276	301	288	45	156	17	334	0	0	23	6		
7	8	10	13	17	19	25	56	94	103	114	142	248	36	200	344	99	188	1	86	341	163	143	249	259	0	0	24	4	11	13			
17	19	22	40	70	88	100	113	143	144	214	302	90	221	3	359	263	295	49	77	260	0	0	25	2	7	12	15	16	17	32			
54	68	99	104	144	145	127	298	45	313	172	44	310	122	188	23	298	0	0	26	3	7	14	16	20	22	38	48	85	98				
100	145	146	210	76	196	27	126	248	72	87	62	138	171	0	0	27	5	9	10	20	21	23	27	52	67	90	94	146	147				
215	238	252	154	354	147	254	222	258	371	0	0	28	2	3	4	7	13	14	16	42	58	82	95	97	147	148	2	13	13	154			
102	123	242	290	126	13	73	198	0	0	29	0	2	9	10	18	21	29	55	69	89	102	148	149	192	26	268	101	49	133				
152	33	161	310	219	0	0	30	1	3	8	9	10	17	20	43	63	76	100	149	150	225	53	256	42	50	265	344	15	72	65			
324	0	0	31	0	2	7	9	21	22	32	47	61	92	116	150	151	251	329	4	144	203	135	90	137	242	215	71	0	0	32	3	8	
12	15	16	19	23	35	64	73	106	151	152	75	9	53	305	219	76	316	278	264	41	194	0	0	33	0	1	5	6	8	9	12	36	
52	72	105	112	152	153	193	289	122	345	268	45	252	142	42	244	261	216	0	0	34	2	3	6	11	13	14	30	53	83				
109	119	153	154	198	238	102	316	125	311	215	304	171	233	25	0	0	35	0	5	7	17	19	23	24	63	78	96	118					
154	155	325	212	281	345	46	101	151	172	236	138	51	0	0	36	2	5	9	10	12	15	19	26	56	66	71	107	155	156				
55	12	237	99	228	150	213	104	91	291	92	131	0	0	37	1	4	6	18	22	23	29	51	61	85	96	156	157	210	300	236			
46	341	2	333	314	107	60	342	0	0	38	2	8	11	14	16	18	19	39	54	73	82	119	157	110	225	136	154	222	159				
50	273	159	158	62	240	76	0	0	39	2	9	13	15	16	20	23	36	49	86	101	158	159	125	324	179	352	157	311	41				
213	260	215	224	0	0	40	1	3	6	12	18	19	31	57	92	95	114	159	160	99	265	202	210	13	265	100	118	275	280				
100	0	0	41	0	4	7	9	13	20	41	50	64	74	96	160	161	266	328	333	343	93	305	316	282	57	341	293	0	0	42	1	4	
14	15	19	20	28	32	86	89	110	161	162	74	103	45	205	151	222	156	106	358	170	294	0	0	43	3	10	11	16	18				
21	22	45	53	58	111	162	163	148	60	91	137	330	224	338	144	128	65	284	0	0	44	2	10	11	12	13	14	15	34				
57	67	68	115	163	21	77	206	51	84	9	316	125	322	96	210	164	4	0	0	45	0	1	9	10	13	20	26	59	80	106	113		
164	165	310	173	346	83	239	206	327	198	262	64	280	0	0	46	2	5	7	8	15	16	19	41	44	70	85	105	165	166				
250	147	233	254	87	92	243	266	13	222	227	290	0	0	47	1	4	6	12	22	23	39	53	55	94	116	166	167	110	124				
212	137	199	102	123	186	133	280	231	0	0	48	0	7	10	12	16	21	22	48	66	87	97	167	168	133	207	254	348					

289 340 189 35 113 273 9 0 49 4 5 10 11 16 23 30 51 62 90 101 168 169 290 18 27 182 317 148 4 19
294 206 311 0 0 50 3 5 14 18 19 22 25 57 89 105 119 169 170 304 298 256 329 99 110 263 181 249 340
13 0 0 51 1 8 9 13 14 16 18 42 71 81 109 170 171 102 90 110 343 80 310 19 327 250 173 93 0 0 52 0 10
11 13 16 20 35 44 86 108 115 171 172 269 356 84 303 323 200 351 49 287 307 200 0 0 53 2 11 17 21 22
23 31 63 93 104 113 172 173 74 127 272 61 163 349 261 183 125 99 117 0 0 54 0 5 6 9 11 13 45 46 65
70 81 173 174 359 30 277 163 179 218 178 286 85 286 327 0 0 55 2 6 8 17 20 23 26 37 62 76 95 174
175 280 8 87 185 136 193 300 25 261 340 120 0 0 56 3 6 8 10 19 21 36 50 60 93 116 175 176 72 89 240
39 250 214 225 22 176 88 291 0 0 57 4 6 7 9 11 12 15 33 68 69 107 176 177 73 65 150 80 133 233 273
228 4 232 6 0 0 58 0 3 7 14 15 17 18 27 46 79 92 118 177 178 124 39 149 116 100 145 240 290 57 304
105 13 0 0 59 1 12 17 18 20 21 30 50 77 99 111 178 179 317 35 148 6 341 32 347 14 326 342 237 0 0 60
3 6 12 14 15 23 34 61 81 87 108 120 179 350 122 72 90 310 344 85 245 10 12 93 0 359 61 1 4 9 10 13
19 38 49 79 83 106 120 121 313 156 2 258 227 245 5 117 280 96 211 0 0 62 3 5 15 20 22 23 44 62 78 91
99 121 122 219 33 282 344 118 250 5 238 352 230 120 0 0 63 0 8 12 16 21 23 28 56 93 98 117 122 123
211 302 339 64 228 247 147 13 38 143 247 0 0 64 1 7 8 9 16 19 20 46 55 64 101 123 124 104 289 301
304 55 48 170 244 39 185 268 0 0 65 2 3 12 13 18 20 24 54 80 112 114 124 125 5 355 305 99 306 88 96
288 22 52 124 0 0 66 1 6 8 14 16 17 34 40 74 102 107 125 126 309 348 241 98 41 83 3 143 140 102 227
0 0 67 4 8 10 12 19 20 35 37 48 75 103 126 127 108 341 333 17 211 187 104 239 140 284 137 0 0 68 0 4
14 17 21 23 33 65 78 82 98 127 128 38 165 219 233 270 282 63 281 264 225 209 0 0 69 5 6 13 18 19 21
22 41 67 76 109 128 129 8 207 200 148 234 33 307 100 203 162 94 0 0 70 1 2 3 5 11 13 14 29 60 91 103
112 129 130 271 291 274 212 110 57 80 180 134 208 244 337 0 0 71 6 7 8 11 15 21 23 38 58 72 74 130
131 253 35 207 278 180 122 120 75 147 0 174 0 0 72 4 11 19 20 21 22 25 47 80 111 115 131 132 124 95
102 48 172 177 244 144 83 207 136 0 0 73 3 5 7 14 15 17 18 37 69 83 84 110 132 133 66 211 233 54
315 342 338 208 229 357 199 143 0 0 74 1 4 7 10 14 15 17 20 28 52 77 88 133 134 43 282 309 252 263
115 220 85 139 214 114 302 0 0 75 0 4 5 6 8 18 39 43 71 79 104 134 135 283 20 166 310 147 229 316
359 268 15 194 0 0 76 1 4 5 18 21 22 40 60 84 97 118 135 136 151 213 230 83 304 132 47 336 276 233
167 0 0 77 0 6 9 14 21 22 23 27 59 75 110 136 137 70 23 169 174 230 15 19 119 25 57 40 0 0 78 1 4 11
18 21 23 31 66 73 77 91 137 138 78 222 317 343 322 29 293 15 97 268 17 0 0 79 3 8 12 17 18 22 47 65
90 102 117 138 139 1 173 0 282 148 282 299 122 42 212 311 0 0 80 0 2 11 13 15 22 24 42 49 75 88 139
140 8 186 278 1 309 186 57 294 255 293 67 0 0 81 0 5 7 9 12 16 17 45 51 84 108 117 140 141 141 342
45 270 83 272 333 19 105 153 314 134 0 0 82 2 5 10 11 15 17 33 43 59 72 87 141 142 280 99 183 227
276 301 288 45 156 17 334 0 0 83 6 7 8 10 13 17 19 25 56 94 103 114 142 248 36 200 344 99 188 1 86
341 163 143 249 259 0 0 84 4 11 13 17 19 22 40 70 88 100 113 143 144 214 302 90 221 3 359 263 295
49 77 260 0 0 85 2 7 12 15 16 17 32 54 68 99 104 144 145 127 298 45 313 172 44 310 122 188 23 298 0
0 86 3 7 14 16 20 22 38 48 85 98 100 145 146 210 76 196 27 126 248 72 87 62 138 171 0 0 87 5 9 10 20
21 23 27 52 67 90 94 146 147 215 238 252 154 354 147 254 222 258 371 0 0 88 2 3 4 7 13 14 16 42 58
82 95 97 147 148 2 13 13 154 102 123 242 290 126 13 73 198 0 0 89 0 2 9 10 18 21 29 55 69 89 102
148 149 192 26 268 101 49 133 152 33 161 310 219 0 0 90 1 3 8 9 10 17 20 43 63 76 100 149 150 225
53 256 42 50 265 344 15 72 65 324 0 0 91 0 2 7 9 21 22 32 47 61 92 116 150 151 251 329 4 144 203 135
90 137 242 215 71 0 0 92 3 8 12 15 16 19 23 35 64 73 106 151 152 75 9 53 305 219 76 316 278 264 41
194 0 0 93 0 1 5 6 8 9 12 36 52 72 105 112 152 153 193 289 122 345 268 45 252 142 42 244 261 216 0 0
94 2 3 6 11 13 14 30 53 83 109 119 153 154 198 238 102 316 125 311 215 304 171 233 25 0 0 95 0 5 7
17 19 23 24 63 78 96 118 154 155 325 212 281 345 46 101 151 172 236 138 51 0 0 96 2 5 9 10 12 15 19
26 56 66 71 107 155 156 55 12 237 99 228 150 213 104 91 291 92 131 0 0 97 1 4 6 18 22 23 29 51 61
85 96 156 157 210 300 236 46 341 2 333 314 107 60 342 0 0 98 2 8 11 14 16 18 19 39 54 73 82 119 157
110 225 136 154 222 159 50 273 159 158 62 240 76 0 0 99 2 9 13 15 16 20 23 36 49 86 101 158 159 125
324 179 352 157 311 41 213 260 215 224 0 0 100 1 3 6 12 18 19 31 57 92 95 114 159 160 99 265 202
210 13 265 100 118 275 280 100 0 0 101 0 4 7 9 13 20 41 50 64 74 96 160 161 266 328 333 343 93 305
316 282 57 341 293 0 0 102 1 4 14 15 19 20 28 32 86 89 110 161 162 74 103 45 205 151 222 156 106
358 170 294 0 0 103 3 10 11 16 18 21 22 45 53 58 111 162 163 148 60 91 137 330 224 338 144 128 65
284 0 0 104 2 10 11 12 13 14 15 34 57 67 68 115 163 21 77 206 51 84 9 316 125 322 96 210 164 4 0 0
105 0 1 9 10 13 20 26 59 80 106 113 164 165 310 173 346 83 239 206 327 198 262 64 280 0 0 106 2 5 7
8 15 16 19 41 44 70 85 105 165 166 250 147 233 254 87 92 243 266 13 222 227 290 0 0 107 1 4 6 12 22
23 39 53 55 94 116 166 167 110 124 212 137 199 102 123 186 133 280 231 0 0 108 0 7 10 12 16 21 22

48 66 87 97 167 168 133 207 254 348 289 340 189 35 113 273 9 0 0 109 4 5 10 11 16 23 30 51 62 90
 101 168 169 290 18 27 182 317 248 4 19 294 206 311 0 0 110 3 5 14 18 19 22 25 57 89 105 119 169 170
 304 298 256 329 99 110 263 181 249 340 13 0 0 111 1 8 9 13 14 16 18 42 71 81 109 170 171 102 90 110
 343 80 310 19 327 250 173 93 0 0 112 0 10 11 13 16 20 35 44 86 108 115 171 172 269 356 84 303 323
 200 351 49 287 307 200 0 0 113 2 11 17 21 22 23 31 63 93 104 113 172 173 74 127 272 61 163 349 261
 183 125 99 117 0 0 114 0 5 6 9 11 13 45 46 65 70 81 173 174 359 30 277 163 179 218 178 286 85 286
 327 0 0 115 2 6 8 17 20 23 26 37 62 76 95 174 175 280 8 87 185 136 193 300 25 261 340 120 0 0 116 3
 6 8 10 19 21 36 50 60 93 116 175 176 72 89 240 39 250 214 225 22 176 88 291 0 0 117 4 6 7 9 11 12 15
 33 68 69 107 176 177 73 65 150 80 133 233 273 228 4 232 6 0 0 118 0 3 7 14 15 17 18 27 46 79 92 118
 177 178 124 39 149 116 100 145 240 290 57 304 105 13 0 0 119 1 12 17 18 20 21 30 50 77 99 111 178
 179 317 35 148 6 341 32 347 14 326 342 237 0 0 Step 5) The parity-check matrix designed based on
 FIG. 4 in step 4 needs to be modified into the form of FIG. 3 by the row permutation and the column
 permutation.

(92) Hereinafter, examples of the parity-check matrix designed by the foregoing method for designing a
 parity-check matrix will be described.

(93) It is well known that any parity-check matrix $H_{\text{sub}.1}$ is equivalent to a parity-check matrix $H_{\text{sub}.2}$
 obtained by performing row permutation and column permutation on the parity-check matrix $H_{\text{sub}.1}$.
 That is, any parity-check matrix having the form of FIG. 3 is equivalent to a parity-check matrix having
 the form of FIG. 4 obtained by performing row permutation and column permutation on the parity-check
 matrix of FIG. 3.

(94) Further, a parity-check matrix having the form of FIG. 3 and a parity-check matrix, in which an
 order of column groups of the parity-check matrix of FIG. 3 according to the exemplary embodiment is
 changed, are also equivalent to each other. Further, when index values of a row at which one (1) is
 positioned in a zero (0)-th column of an i -th column group are $k_{\text{sub}.1}, k_{\text{sub}.2}, \dots, K_{\text{sub}.m}$ (herein, m
 is a column degree of the i -th column group), parity-check matrices are equivalent to each other even
 when the index values of the row at which one (1) is positioned in the zero (0)-th column of the i -th
 column group are changed to $(K_{\text{sub}.x} + Q_{\text{sub}.ldpc} \times y) \bmod (N_{\text{sub}.ldpc} - K_{\text{sub}.ldpc})$ ($1 \leq k \leq m$).
 $Q_{\text{sub}.ldpc}$ is $(N_{\text{sub}.ldpc} - K_{\text{sub}.ldpc) / M}$ as described above, in which M represents the number of
 columns belonging to a column group. Further, y is any constant.

(95) As a specific example, in the case in which $N_{\text{sub}.ldpc}$ is 30, $K_{\text{sub}.ldpc}$ is 15, and $Q_{\text{sub}.ldpc}$ is 3, if
 a case in which position information about rows at which weight-1s are positioned in zero (0)-th columns
 of three column groups is represented as following Table 6,

(96) TABLE-US-00006 TABLE 6 1 2 8 10 0 9 13 0 14

(97) When y is 2,
 since $1.\text{fwdarw.}(1+3 \times 2) \bmod 15 = 7$
 $2.\text{fwdarw.}(2+3 \times 2) \bmod 15 = 8$
 $8.\text{fwdarw.}(8+3 \times 2) \bmod 15 = 14$
 $10.\text{fwdarw.}(10+3 \times 2) \bmod 15 = 1$
 $0.\text{fwdarw.}(0+3 \times 2) \bmod 15 = 6$
 $9.\text{fwdarw.}(9+3 \times 2) \bmod 15 = 0$
 $13.\text{fwdarw.}(13+3 \times 2) \bmod 15 = 4$
 $0.\text{fwdarw.}(0+3 \times 2) \bmod 15 = 6$
 $14.\text{fwdarw.}(14+3 \times 2) \bmod 15 = 5,$

(98) the parity-check matrices are equivalent to one another even when the position information about
 the row at which the weight-1s are positioned in the zero (0)-th columns of the three column groups is
 represented as following Table 7.

(99) TABLE-US-00007 TABLE 7 7 8 14 1 6 0 4 6 5

(100) Further, the parity-check matrices are equivalent to one another even when an order of the column
 groups is changed, and thus, the parity-check matrices represented as the following Table 8 are also
 equivalent to one another.

(101) TABLE-US-00008 TABLE 8 0 9 3 0 14 1 2 3 10

(102) Further, in the case of the parity-check matrix having the form of FIG. 4 according to the
 exemplary embodiment, when the index values of the quasi-cycle matrix present in the same column

group or column block are added or subtracted to or from the same value, the parity-check matrix may be equal to the previous parity-check matrix. Further, the parity-check matrices in which an order of the column-blocks is changed are equivalent to one another.

(103) Hereinafter, examples of a parity-check matrix having the form of FIG. 3 designed by the method for designing a parity-check matrix described in the exemplary embodiment will be described.

(104) As an example, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is 6/15, M is 360 and $Q_{\text{sub.ldpc}}$ is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 9.

(105) TABLE-US-00009 TABLE 9 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0

891	2309	3615	5472	8525	10308	11096	11503	11553	12420	13298	15489	15883	17561
21293	23132	25133	28163	30811	31810	34215	34584	36541	38410	1	3290	5479	5596
6546	7021	7566	9274	9303	10047	10278	10956	12393	18309	19755	19790	20828	22690
25235	26853	31571	31893	36500	36772	37536	2	1031	4765	5011	5200	6189	8985
12056	12366	14439	15482	16521	18338	19165	19476	21455	22096	22589	23699	23838	24649
30113	31490	31960	34564	3	1619	2364	2889	3409	7278	7384	8220	11204	11694
12064	16087	16722	19633	21525	24373	25817	29614	31392	33508	34329	34352	37006	37080
38444	4	3612	5474	6202	8521	11153	12202	12213	12962	13868	14359	14385	15072
15375	18156	18308	26375	28004	28905	29697	30000	30686	32819	34407	38688	5	1441
2527	3103	3297	6303	6421	7233	8011	8250	8909	10558	13855	14699	17508	17810
21259	22423	23264	29249	30830	32607	35202	38303	38402	6	4	515	1864	6876
7245	7371	9190	11814	13748	15774	20161	21526	24462	24884	25737	26251	27266	29957
34779	35082	36664	37354	37516	37837	7	368	4867	6985	7663	15368	15947	16235
16544	17468	18936	21544	21903	22422	23080	25411	25938	26128	29554	29847	31845	32714
36101	37324	38004	8	1125	2843	3161	3172	6296	8216	9506	9822	10005	11459
12555	12919	13814	13872	17916	19481	22046	23394	26020	32002	35467	36616	38306	38784
9	803	1133	1439	4618	5708	7048	8114	8195	9089	10238	12403	12520	14005
14194	16527	19186	20343	22595	27314	30831	35338	36223	36754	38504	10	988	1723
2736	2849	3827	7076	12613	12989	13268	14323	15013	20703	20872	21170	25384	29187
29524	29693	30029	30948	32598	32733	32906	34586	11	6085	8070	8611	10860	11848
13126	15737	19744	21503	21572	21598	22468	23444	23753	26696	26722	29679	29999	31803
32206	33479	34221	36608	38731	12	1410	1790	3290	7470	9833	10001	10966	11262
11687	12155	12807	16606	17295	20105	23354	23405	27107	28277	29386	30326	32151	32641
33649	36765	13	817	2781	2937	3886	5881	9657	10675	10816	11934	13044	16160
19321	19567	24463	26365	28182	29386	31428	31708	33113	33975	34629	36265	36650	14
1393	6471	7153	7266	7647	11124	13430	16396	21033	21227	23273	24502	25273	27888
29107	30726	31551	31987	32737	33126	33764	35951	36021	36264	15	5003	10529	15986
32408	16	1236	7663	14599	24745	17	10824	16168	16783	29318	18	6834	29474
29608	29779	19	2709	8594	20859	23739	20	6014	11516	14715	36702	21	2784
6493	27798	35084	22	1044	5892	13056	18720	23	2875	19737	21893	32447	24
11358	17602	34706	38472	25	11527	23325	28196	33428	26	2428	2597	3177	3909
27	29269	33353	33791	37487	28	204	14925	22290	26379	29	18959	19007	24212
27094	30	13813	14631	17097	31294	31	11533	17559	30617	37966	32	14821	22942
28281	28353	33	3440	8417	8534	24133	34	10186	20333	24737	34733	35	12388
31132	33131	36927	36	7311	16914	19363	30928	37	16762	23990	30952	36068	38
1872	9840	33270	37616	39	14047	17094	18244	24107	40	1903	21121	22616	38860
41	5979	16278	27327	34338	42	7644	10438	25622	36926	43	1243	27018	31393
38210	44	179	1944	5719	15319	45	17318	24311	37325	46	1528	10135	33640
47	1321	15978	25776	48	16957	21572	26719	49	14349	15799	29164	50	574
5582	20145	51	8814	14785	31081	52	667	7530	25659	53	7823	23615	37483
54	12766	30755	35696	55	4302	11660	36217	56	7393	11724	20441	57	864
13910	22924	58	5500	10085	31057	59	5399	13946	32583	60	1364	12423	36203
61	12356	24958	32026	62	10894	33813	38525	63	15456	15794	28350	64	6603
25570	33797	65	8709	19457	38831	66	330	9502	24362	67	12579	15733	34408
68	28443	34954	37293	69	3123	8513	19702	70	16976	22876	26384	71	20399
29971	33952												

(106) That is, positions of a row at which one (1) is positioned in a zero (0)-th column of each column group may be defined by indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group shown in above Table 9, and, by shifting a row at which one (1) is positioned in a zero (0)-th column of each column group by $Q_{\text{sub.ldpc}}$, rows at which one (1) is positioned in other columns of the column group may be defined.

(107) In the foregoing example, since $Q_{\text{sub.ldpc}} = (64800 - 25920) / 360 = 108$ and the indices of the row at

(111) As another example, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, the code rate R is 10/15, M is 360 and $Q_{\text{sub.ldpc}}$ is 60, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 11.

(112) TABLE-US-00011 TABLE 11 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0

114	2135	3045	4635	5512	5681	6571	8943	10053	10109	13161	13668	14218	17417
19328	21140	1	2639	2821	3066	4293	5350	8130	9037	11265	12556	15047	15364
15531	15700	16938	17202	19034	2	1891	4150	4822	4855	6646	9754	10460	14005
14139	15038	17213	18336	20069	20384	21305	21508	3	305	600	3410	5170	5740
7354	8462	9026	12763	17132	17336	17653	18450	19318	20848	21559	4	1961	3637
4249	4694	8298	8784	8836	11708	12241	14172	14207	15127	15462	17277	20415	20848
5	1101	3770	7816	8727	8890	8915	8953	11655	12826	14313	15682	19622	19854
20569	20916	21129	6	726	933	3015	5034	6431	6743	7477	8927	9189	9520
14280	15514	16316	17757	20237	21175	7	1661	3074	3745	4264	4775	7633	7666
9228	12388	12657	12718	17066	18921	19463	19511	21391	8	1147	3483	3544	5553
6270	6406	7146	7256	8138	9191	9623	11239	12795	16251	16435	21092	9	885
1061	2199	3364	5421	5549	7347	7416	11477	11874	12991	15051	16857	18933	19110
21481	10	292	983	1627	6121	6408	6494	6507	10642	15569	15696	16665	17024
18043	18630	19316	20029	11	2598	2674	3504	4931	4940	8002	9284	10729	10914
13478	13677	14033	15010	15912	16183	16612	12	19	768	1263	3305	6513	7677
7956	9040	13427	16641	17280	18452	18584	18925	19559	20587	13	1071	3472	7305
7981	8574	9609	10899	14134	15508	15665	15683	16061	16224	16604	18190	21560	14
2974	5834	6290	8468	9866	11177	12398	14248	14698	15726	16200	16810	16851	18373
18942	21104	15	519	2684	2713	2845	3000	3080	3332	4682	5062	5277	9342
10811	12636	14714	15658	16426	16	2272	2629	3051	4308	5301	7108	8318	8492
11305	12219	13423	16126	17763	18304	19146	20006	17	935	1093	1641	3562	4699
5333	5730	7628	8364	8414	10343	10555	12779	12958	16626	18985	18	1038	1333
1843	1910	3245	7258	7875	12098	12729	12739	16636	18689	18877	20511	20860	21299
19	5740	6656	6901	7066	7569	8856	8947	12582	15492	15710	17072	18638	18724
18875	21444	21563	20	962	990	1199	2979	3341	8322	9285	9652	10387	11404
12387	13495	14066	16325	16514	18732	21	387	1248	2298	3376	5408	7817	7923
8203	8816	9451	11292	13649	14291	17993	19629	19739	22	84	1177	1363	3189
4699	6746	9707	10308	10460	10992	11873	13531	13696	14522	15050	20717	23	713
960	2672	4688	6602	6769	6783	10075	12807	14411	15527	15575	19179	19878	20477
21517	24	12575	15845	18200	25	5870	6972	16463	26	2025	3655	15396	27
4258	6387	14477	28	12282	12783	13274	29	1657	10810	12509	30	839	8734
21409	31	4038	5993	15640	32	3025	15282	16231	33	4342	7977	17828	34
14144	16500	21426	35	592	4952	15367	36	8156	8859	13113	37	7267	9133
20155	38	17111	17306	21301	39	2655	5258	14267	40	5844	13026	18796	41
2681	5686	15609	42	2031	3980	4228	43	75	18922	20730	44	18712	20866
21302	45	10974	13003	20481	46	4494	6964	18238	47	3679	12972	13411	48
13207	16406	19548	49	6039	6320	14581	50	4721	20336	20819	51	2797	15321
20509	52	8307	8774	19113	53	3394	10487	13963	54	4325	12098	14305	55
13667	19264	19649	56	1163	16176	20823	57	2324	10790	14560	58	12791	14068
17743	59	9765	12262	20117	60	1456	11096	13570	61	6900	7111	15217	62
4009	5995	7322	63	10673	11315	17310	64	5792	10504	18221	65	4748	14299
16554	66	4176	14868	20718	67	6147	9429	15884	68	9044	10345	21417	69
7737	7873	11969	70	3924	4494	8326	71	5535	6651	16116	72	11	6993
20602	73	15798	17918	19172	74	1181	11171	13206	75	4040	4567	18197	76
1255	11889	17730	77	2099	5538	14774	78	482	5768	7475	79	3418	4801
20715	80	5925	16632	20285	81	2034	11271	21000	82	7238	8108	20848	83
193	11374	15841	84	5056	9673	12441	85	8026	17906	18037	86	162	4432
8739	87	1582	5268	20880	88	3494	17600	18684	89	3029	6710	11442	90
9289	19099	19407	91	7802	9130	20598	92	5140	8731	15358	93	14153	16376
19323	94	4847	11843	21567	95	4840	14455	17248	96	1117	4061	13355	97
7636	9748	21108	98	9068	13023	13346	99	1139	14402	20245	100	2190	11366
17004	101	2989	5524	8199	102	8489	8899	15486	103	6683	6970	13387	104
3745	9975	15713	105	1250	4246	5973	106	8941	9992	17805	107	7986	13776
21297	108	2781	3232	16020	109	7654	15969	16071	110	4002	13033	19217	111
4603	7439	9192	112	1390	8673	18485	113	4845	6024	14633	114	6083	14165
15640	115	9652	13452	21404	116	4196	7787	17371	117	2959	6783	13581	118
11596	18575	20878	119	17078	20134	20870							

(113) As another example, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, the code rate R is 12/15, M is 360 and $Q_{\text{sub.ldpc}}$ is 36, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 12.

(114) TABLE-US-00012 TABLE 12 i Indices of rows at which 1s are positioned in 0-th column of i -th

column group 0 57 3127 3324 3362 3471 4108 4850 5588 5721 6436 6845 9325 11745 12956 1 651 893
1879 2641 2807 4123 4406 5364 6346 8530 9387 10301 11047 12660 2 1925 2592 2655 4108 4433
4544 4614 7081 7254 7759 7815 9121 9453 10613 3 718 1559 1859 2697 3739 3948 4506 4982 6898
7410 7569 8243 10840 12075 4 21 1387 1645 1980 4188 4751 6148 7864 8498 8647 9715 10513 11114
12910 5 1082 2045 2395 3663 4597 4727 7055 8123 8189 8618 8782 11114 11716 12211 6 312 951
2269 2569 4820 4972 5803 5816 6406 6861 8277 8398 9349 10816 7 20 1833 2300 4962 5267 5653
6215 7098 7434 7509 9235 10331 10731 11060 8 594 1091 1743 1809 2686 3095 3316 5381 5927 6115
7100 7750 9901 11091 9 1065 1236 1716 2628 4589 5164 5209 6502 7719 10107 11333 11599 12011
12893 10 1799 2958 3999 4196 4351 4358 5019 5128 5840 6154 6961 9328 10530 11440 11 2982 3133
4612 4776 4986 6276 6992 7372 7589 7831 8109 8619 11594 12694 12 282 3140 3300 4219 4459 5189
5525 6753 8154 8522 8931 9073 10019 10654 13 298 1137 3973 4000 7328 7923 8602 8734 8792 9419
9748 9988 10200 10985 14 1142 1639 1997 3951 4388 4638 5676 6478 6738 6927 8643 8793 11528
11949 15 629 1735 3072 3268 3585 3888 4491 4622 5294 6817 6829 9003 9930 10426 16 486 1200
1594 2713 3273 3440 4667 5247 7304 7901 8302 8576 9014 10180 17 2377 2471 3742 5186 7137 9518
9798 10029 10060 11052 12109 12660 12716 12762 18 175 1242 2118 4101 4130 6252 6598 7272 7342
7834 7859 7924 10940 11032 19 550 1349 2360 2373 2895 3442 5258 5695 8512 8882 10710 11562
11940 12432 20 1791 2788 4958 5573 5917 5952 6195 7376 7920 7971 8983 10817 10883 11550 21
639 1000 2387 2429 3548 3659 4186 6741 7490 8383 8760 9392 9649 11971 22 523 1269 2262 2987
3073 3811 4227 5028 5181 8471 8712 9421 9509 11871 23 117 266 505 1967 2620 3188 5282 7194
8119 8251 8290 10433 12174 12337 24 896 3196 3714 4979 5258 5584 5945 7142 7713 9745 9789
11252 11339 11945 25 864 2383 2594 3398 5969 6116 6418 6649 7199 7242 8169 9221 9740 9890 26
1334 1487 12881 27 1447 4633 11032 28 273 1257 10453 29 5102 10999 12496 30 7578 9743 10513 31
7156 10266 10546 32 3199 7369 7824 33 2354 7674 8569 34 3687 9478 12663 35 2997 9418 9581 36
6519 8081 11229 37 9894 9899 11254 38 6 9344 11731 39 9612 12680 12776 40 353 10896 11643 41
945 2537 9804 42 2915 8984 11098 43 2353 4884 7456 44 6003 8924 11646 45 349 748 8625 46 4799
7204 12240 47 2464 8958 11020 48 1915 2903 12358 49 2246 3032 12531 50 2594 12742 12914 51
2002 10995 12079 52 853 1049 5022 53 4142 4301 6413 54 914 3882 12047 55 5479 10413 11225 56
228 6874 11183 57 2836 9737 10728 58 1795 9981 12734 59 2641 2844 11779 60 1245 9983 12804 61
6002 8612 9704 62 1237 1760 7504 63 844 1485 5869 64 2657 4461 5642 65 2423 4203 9111 66 244
1855 6131 67 5318 6371 11430 68 391 1617 10126 69 1762 9259 10603 70 2604 4335 6702 71 4381
5486 8045 72 7667 8875 11451 73 1968 4023 6911 74 4630 10184 11357 75 6582 12348 12769 76 3840
6302 7388 77 1 4197 5358 78 1265 3153 11352 79 2504 7180 10044 80 1980 5027 9717 81 5699 6899
9668 82 1432 2803 3314 83 1237 8470 9642 84 829 6745 7923 85 329 1931 5575 86 1067 6867 7257
87 4744 5559 7835 88 109 6756 12238 89 2814 5237 11153 90 1592 10696 10749 91 225 2293 6260 92
646 2170 7578 93 6466 8222 9838 94 44 6574 12160 95 1755 2734 12780 96 1249 8264 8318 97 5789
6622 9481 98 5666 8681 8744 99 123 5803 9291 100 3750 7919 9167 101 1064 2848 12753 102 2142
8656 9244 103 6193 7219 11732 104 7356 7819 9928 105 4780 5937 11993 106 5092 7186 9141 107
1238 3840 12360 108 1649 2096 2587 109 5560 5903 12899 110 1134 4341 8330 111 1645 9495 10041
112 9585 11595 12912 113 8748 10646 11894 114 45 7255 9074 115 1051 2694 6188 116 622 3460
8394 117 3598 4623 9025 118 1218 3540 12843 119 4938 8698 12423 120 1766 3635 11427 121 5177
6706 9127 122 943 3590 10245 123 4864 7394 11117 124 5852 6042 10421 125 8285 10775 12349 126
787 7171 7866 127 718 4688 12234 128 728 2353 10667 129 3629 4592 6485 130 2880 5157 11466 131
2906 10220 11796 132 4243 5440 10907 133 5262 7543 12303 134 4440 7779 10940 135 2515 5843
9202 136 4684 8874 10586 137 2270 7197 8652 138 7190 7870 8317 139 1158 10456 12909 140 1583
7669 10781 141 8141 11209 12555 142 3181 3903 7832 143 2428 4467 8074

(115) Meanwhile, in the design process of codes, the process of step 4 uses the lifting method considering the cycle characteristics. When various algebraic characteristics as well as the cycle characteristics are additionally considered, codes having better performance may also be designed.

(116) Generally, since the performance of the LDPC codes affects the degree distribution as well as the cycle characteristics, codes having better performance may be designed in consideration of both of the two characteristics.

(117) According to an exemplary embodiment, parameters may be determined depending on cycles having the shortest length and the number of variable nodes included in these cycles and having a

specific degree, and a sequence may be determined depending on a rule determined based on the parameters in a lifting step.

(118) For example, when the lifting process is applied to a column group of which the degrees are A and B, only a cycle in which the number of variable nodes of which the degree is A among the variable nodes included in the cycle while the lifting of the column group of which the degree is A is performed is $x_{\text{sub.1}}$ or less and a value of the variable nodes (the number of variable nodes of which the degree is $A + \text{the number of variable nodes of which the degree is } B \times C$) included in the cycle while the lifting of the column group of which the degree is B is $x_{\text{sub.2}}$ or less may be considered.

(119) However, all cycles are considered when the lifting process is applied to the column groups of which the degrees are not A and B.

(120) Here, A is a positive integer of two (2) or more, B is a positive integer larger than A and smaller than a maximum degree of the parity-check matrix, and C is a weighting factor and has a positive integer value.

(121) In addition, all cycles are considered without distinction of a specific cycle when the lifting process is applied to the column groups of which the degrees are not A and B.

(122) As a specific example, the case in which $A=3$, $B=4$, and $C=2$ will be described.

(123) For example, in the lifting process, the lifting is performed in consideration of only cycles in which the number of variable nodes (which is equivalent to columns) of which the degree is three (3) among the variable nodes included in the cycles in the process of performing the lifting of column groups of which the degrees are three (3) is $x_{\text{sub.1}}$ or less and (the number of variable nodes of which the degree is $4 \times 2 + \text{the number of variable nodes of which the degree is three (3)}$) among variable nodes included in the cycles in the process of performing the lifting of column groups of which the degree is four (4) is $x_{\text{sub.2}}$ or less.

(124) That is, when the index values are changed in the lifting process, the lifting process is repeated until the number of variable nodes having the cycles is not improved to change the index values depending on the degrees of each variable node in consideration of only the cycles corresponding to the conditions.

(125) Generally, in simple lifting, a parity-check matrix is designed so that the number of cycles is decreased. In this process, the parity-check matrix is designed such that the number of cycles meeting the above-mentioned conditions is decreased.

(126) The reason why the above-mentioned conditions affect the design of LDPC codes having better performance is that the cycles and the degree distributions affecting the LDPC codes have been considered in the above-mentioned conditions.

(127) Examples of the LDPC codes designed in consideration of these additional conditions are shown in following Tables 13 to 18. For reference, the same length and code rate as those of the LDPC codes as shown in above Tables 9 to 12 are designed, values ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 13 are equal to (4, 5), values ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 14 is equal to (4, 5), ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 15 are equal to (3, 4), and values ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 16 are equal to (3, 4). Further, values ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 17 are equal to (8, 8) and ($x_{\text{sub.1}}$, $x_{\text{sub.2}}$) considered in following Table 18 is equal to (3, 4).

(128) As an example of results designed in consideration of these additional conditions, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is 6/15, M is 360, and $Q_{\text{sub.ldpc}}$ is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 13.

(129) TABLE-US-00013 TABLE 13 i Indices of rows at which 1s are positioned in 0-th column of i-th column group 0 253 1553 2024 4493 5350 5664 6351 8563 8803 9576 9936 11995 12166 13193 13412 17088 21893 22167 22478 22673 23735 35313 36981 37527 1 1238 1701 2316 3621 7222 8387 9190 10178 10599 10744 11743 18179 22465 23562 23820 26493 28168 28515 28940 29982 31196 32946 33687 34305 2 1475 1874 2994 3168 3381 3988 7646 9309 11199 12856 15025 16586 20113 23155 24030 27491 28235 29392 30885 32896 33656 33785 34685 38713 3 512 721 1813 3144 3276 6198 13540 14553 15017 15548 16120 17449 19101 19763 24180 25629 27612 29682 30910 32038 35346 36607 36836 37876 4 333 1124 3269 4236 4920 5207 6154 7041 7282 7979 8472 9302 10033 14295 15888 16485 18963 24572 26642 27516 30242 31209 32000 33259 5 2528 4425 4656 6631 8875 10621

12590 14334 14011 16406 16937 18942 20315 24078 24889 27298 29555 30123 30513 33101 33403
 33787 36651 37975 6 414 721 898 2093 3813 8358 9316 11235 12032 12568 14339 14908 15390 19210
 19450 22689 22840 28044 31218 35042 35348 35863 37611 37837 7 979 3559 4988 6900 7254 7491
 11518 12297 12928 14894 15473 16179 16667 17744 20983 22854 24913 25640 28792 29536 30428
 32284 33732 34523 8 353 1877 2171 5080 7140 7878 8762 11558 14836 15000 15513 16490 17423
 18943 21094 22348 23394 24182 26203 28328 32408 34284 38061 38259 9 142 1961 5219 5816 9555
 10358 10675 15251 15716 16079 20566 21470 23007 25419 27130 28000 28693 30742 30862 33209
 33472 33851 35546 36115 10 1388 4346 5764 6052 7940 9207 13401 13603 14233 14411 16310 16598
 22524 22824 26535 27965 28433 30605 31984 34241 35730 36139 37261 37720 11 106 1441 3892
 4300 5026 7207 8648 14012 15828 17007 19409 22942 26363 28363 28806 29351 29722 33033 35204
 35315 35824 37901 38036 38643 12 2607 5309 6506 9122 9318 9889 10322 12074 13373 14058 15341
 15774 18154 18749 21949 27887 29885 31294 31487 32769 32890 37983 38403 38879 13 2154 4374
 10366 10605 11179 15994 18855 20342 23936 24777 25768 26371 26745 27049 27324 27493 27985
 29781 30148 33240 33673 34441 35057 35731 14 3054 6385 6561 6755 7795 9366 10392 12042 12832
 14851 16187 17441 18536 20967 21792 24084 24505 26677 28167 28334 35199 36745 37533 37786
 15 2735 3833 10268 17606 16 559 9607 17652 34573 17 9528 12139 14306 38416 18 11978 17094
 25891 30040 19 14811 15531 27333 31274 20 12926 28602 32103 35600 21 8616 14917 18992 23478
 22 7416 8568 10248 36672 23 3611 15521 17793 32467 24 4662 7908 8894 32722 25 8156 28077
 32840 35719 26 977 17949 23380 26181 27 7655 9515 12185 17821 28 1107 8520 16782 23241 29
 16942 19784 22031 22955 30 1054 23163 23793 37789 31 8509 17981 22306 33327 32 2253 11397
 16225 30826 33 23654 30073 31421 33140 34 18286 19445 29945 38405 35 7535 8608 21915 38260
 36 11082 13972 17895 18931 37 8114 25376 25660 33718 38 3380 5946 20124 37272 39 10159 18822
 22456 29291 40 15289 18016 18512 37003 41 18471 19326 27834 28119 42 15326 28272 36142 37826
 43 12277 20646 22411 22658 44 12199 23328 35495 36055 45 17318 24311 29009 46 1528 32815
 33640 47 15978 19141 25776 48 8683 11960 16957 49 15799 20829 29164 50 574 5582 20145 51 8814
 18673 31081 52 8394 25659 36739 53 23615 32879 37483 54 12766 30755 35696 55 4302 18788
 25093 56 13205 14424 33529 57 864 16070 22924 58 7120 10085 31057 59 5399 13946 32583 60 1364
 12423 19355 61 12356 24958 32026 62 11997 25582 36581 63 6074 9516 18414 64 7102 22587 33797
 65 8709 19457 38831 66 9502 10266 24362 67 1801 12579 34408 68 471 34954 37293 69 1666 3123
 8513 70 9160 11576 26384 71 14311 20399 33952

(130) As another example of results designed in consideration of these additional conditions, if codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is $8/15$, M is 360, and $Q_{\text{sub.ldpc}}$ is 84, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 14.

(131) TABLE-US-00014 TABLE 14 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0 2768 3039 4059 5856 6245 7013 8157 9341 9802 10470 11521 12083 16610 18361 20321 24601 27420 28206 29788 1 2739 8244 8891 9157 12624 12973 15534 16622 16919 18402 18780 19854 20220 20543 22306 25540 27478 27678 28053 2 1727 2268 6246 7815 9010 9556 10134 10472 11389 14599 15719 16204 17342 17666 18850 22058 25579 25860 29207 3 28 1346 3721 5565 7019 9240 12355 13109 14800 16040 16839 17369 17631 19357 19473 19891 20381 23911 29683 4 869 2450 4386 5316 6160 7107 10362 11132 11271 13149 16397 16532 17113 19894 22043 22784 27383 28615 28804 5 508 4292 5831 8559 10044 10412 11283 14810 15888 17243 17538 19903 20528 22090 22652 27235 27384 28208 28485 6 389 2248 5840 6043 7000 9054 11075 11760 12217 12565 13587 15403 19422 19528 21493 25142 27777 28566 28702 7 1015 2002 5764 6777 9346 9629 11039 11153 12690 13068 13990 16841 17702 20021 24106 26300 29332 30081 30196 8 1480 3084 3467 4401 4798 5187 7851 11368 12323 14325 14546 16360 17158 18010 21333 25612 26556 26906 27005 9 6925 8876 12392 14529 15253 15437 19226 19950 20321 23021 23651 24393 24653 26668 27205 28269 28529 29041 29292 10 2547 3404 3538 4666 5126 5468 7695 8799 14732 15072 15881 17410 18971 19609 19717 22150 24941 27908 29018 11 888 1581 2311 5511 7218 9107 10454 12252 13662 15714 15894 17025 18671 24304 25316 25556 28489 28977 29212 12 1047 1494 1718 4645 5030 6811 7868 8146 10611 15767 17682 18391 22614 23021 23763 25478 26491 29088 29757 13 59 1781 1900 3814 4121 8044 8906 9175 11156 14841 15789 16033 16755 17292 18550 19310 22505 29567 29850 14 1952 3057 4399 9476 10171 10769 11335 11569 15002 19501 20621 22642 23452 24360 25109

25290 25828 28505 29122 15 2895 3070 3437 4764 4905 6670 9244 11845 13352 13573 13975 14600
15871 17996 19672 20079 20579 25327 27958 16 612 1528 2004 4244 4599 4926 5843 7684 10122
10443 12267 14368 18413 19058 22985 24257 26202 26596 27899 17 1361 2195 4146 6708 7158 7538
9138 9998 14862 15359 16076 18925 21401 21573 22503 24146 24247 27778 29312 18 5229 6235
7134 7655 9139 13527 15408 16058 16705 18320 19909 20901 22238 22437 23654 25131 27550
28247 29903 19 697 2035 4887 5275 6909 9166 11805 15338 16381 18403 20425 20688 21547 24590
25171 26726 28848 29224 29412 20 5379 17329 22659 23062 21 11814 14759 22329 22936 22 2423
2811 10296 12727 23 8460 15260 16769 17290 24 14191 14608 29536 30187 25 7103 10069 20111
22850 26 4285 15413 26448 29069 27 548 2137 9189 10928 28 4581 7077 23382 23949 29 3942 17248
19486 27922 30 8668 10230 16922 26678 31 6158 9980 13788 28198 32 12422 16076 24206 29887 33
8778 10649 18747 22111 34 21029 22677 27150 28980 35 7918 15423 27672 27803 36 5927 18086
23525 37 3397 15058 30224 38 24016 25880 26268 39 1096 4775 7912 40 3259 17301 20802 41 129
8396 15132 42 17825 28119 28676 43 2343 8382 28840 44 3907 18374 20939 45 1132 1290 8786 46
1481 4710 28846 47 2185 3705 26834 48 5496 15681 21854 49 12697 13407 22178 50 12788 21227
22894 51 629 2854 6232 52 2289 18227 27458 53 7593 21935 23001 54 3836 7081 12282 55 7925
18440 23135 56 497 6342 9717 57 11199 22046 30067 58 12572 28045 28990 59 1240 2023 10933 60
19566 20629 25186 61 6442 13303 28813 62 4765 10572 16180 63 552 19301 24286 64 6782 18480
21383 65 11267 12288 15758 66 771 5652 15531 67 16131 20047 25649 68 13227 23035 24450 69
4839 13467 27488 70 2852 4677 22993 71 2504 28116 29524 72 12518 17374 24267 73 1222 11859
27922 74 9660 17286 18261 75 232 11296 29978 76 9750 11165 16295 77 4894 9505 23622 78 10861
11980 14110 79 2128 15883 22836 80 6274 17243 21989 81 10866 13202 22517 82 11159 16111 21608
83 3719 18787 22100 84 1756 2020 23901 85 20913 29473 30103 86 2729 15091 26976 87 4410 8217
12963 88 5395 24564 28235 89 3859 17909 23051 90 5733 26005 29797 91 1935 3492 29773 92 11903
21380 29914 93 6091 10469 29997 94 2895 8930 15594 95 1827 10028 20070

(132) As another example of results designed in consideration of these additional conditions, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is $10/15$, M is 360, and $Q_{\text{sub.ldpc}}$ is indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table

(133) TABLE-US-00015 TABLE 15 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0 501 1533 5943 9232 10258 10428 10965 11934 14081 14708 15509 17251 18380
19815 20075 20237 1 753 1266 2017 3107 4210 4770 6520 10861 11594 13191 14116 18342 18604
18825 19398 21479 2 2379 3029 3140 3398 4528 6562 8575 9593 11196 11585 12931 13064 13825
15886 20854 21010 3 733 1262 2250 4910 8165 8374 8698 10543 10930 12940 14520 14936 15752
16879 19226 20188 4 2056 2341 4237 4807 6469 7708 8895 9548 13274 13404 13481 14082 15647
17712 19377 19638 5 2882 3081 3633 4047 4755 5094 5589 6709 12526 12710 12910 13342 14196
17836 20353 21095 6 34 309 1187 3000 3097 3246 6280 6873 7074 8935 12615 13517 14363 16317
19856 20591 7 2181 2381 3551 6904 6995 8248 9023 9348 9433 11097 12914 17326 17671 18658
19585 20846 8 915 2036 2104 2790 3606 4763 6319 7807 8918 9311 13431 15723 19953 20935 21092
21286 9 1921 2131 2321 3114 4589 5133 5477 8265 9891 13941 14404 15777 17310 17787 18399
20916 10 1612 3502 3696 5084 6421 7410 7723 8467 8787 9434 11516 14329 14505 17723 19229
19308 11 2900 3311 3430 3984 4843 5422 6049 7374 7572 11037 15112 15173 17144 20378 20718
20854 12 4685 4896 7712 9120 10019 11988 12657 12907 13113 13519 14384 15347 16180 17125
19923 20661 13 189 2543 5548 6001 8979 9224 12641 13404 13505 13674 16011 16234 16820 17230
18945 19912 14 788 6540 7724 9898 13150 13817 15313 15554 15928 16118 17734 18170 18386
19422 20106 20931 15 4779 4964 6722 8474 9298 10620 11326 11471 12897 13482 15805 17076
18193 18260 20122 21392 16 385 2524 3486 4503 6708 7712 8632 8908 9283 10826 12081 12699
16551 16846 19058 20749 17 1405 1748 3058 4219 9053 9906 10581 11242 11515 14910 15143 16499
18395 18853 19454 21264 18 1875 2716 5358 6878 7089 8758 9659 11909 12290 12697 14631 17200
18305 18973 19159 21583 19 683 1564 1718 3350 3940 5672 6189 9361 11347 11915 13236 15946
16404 18372 21116 21282 20 962 4602 5035 5827 8007 10139 11524 11970 13479 13586 16061 18532
18705 19152 19625 20054 21 497 1698 1976 2383 2823 3479 7527 9948 11889 13649 14491 15431
16868 17232 18316 20453 22 1029 6199 8477 9707 10400 10913 11617 11923 12482 14690 14988
15796 16104 19272 20426 20731 23 95 953 1208 1818 5640 5797 8852 9399 9595 10877 11087 13129

15122 16631 17643 18674 18877 24 12575 12680 15845 25 5870 6972 26 2025 3655 7716 27 4258
6387 14477 28 12282 13274 18603 29 1657 10810 12509 30 839 8734 21409 31 4038 5993 15640 32
14005 15282 18931 33 4342 7977 17828 34 14144 16500 21426 35 592 4952 13747 36 8156 8859
18093 37 5995 7267 9133 38 5581 17111 17306 39 3218 7635 14267 40 5844 13026 18796 41 5686
12821 15609 42 3980 4228 7731 43 75 18922 20730 44 18712 20866 21302 45 10674 13003 20481 46
1914 6964 18238 47 12972 13411 21559 48 13207 16406 19548 49 4021 6039 6320 50 4721 20336
20819 51 2797 15321 20509 52 8187 8774 19113 53 3634 10487 13963 54 3205 4325 12098 55 1444
4409 13667 56 1163 5856 7623 57 2324 10790 14560 58 1468 12791 17743 59 9765 12262 20117 60
1456 4150 11096 61 6900 9631 15217 62 4009 5995 7322 63 10673 11315 17310 64 5792 10504 18221
65 4748 14299 14994 66 4176 14868 20718 67 6147 15884 19749 68 9044 10345 20757 69 5817 7873
11969 70 3114 3924 8326 71 5535 6651 16116 72 7642 16391 17673 73 15798 17918 19172 74 1181
5291 17166 75 4220 4567 18197 76 1255 17730 19449 77 2099 5538 14774 78 6542 7475 17228 79
3418 4801 20715 80 1145 4245 16632 81 2034 11271 21000 82 7238 8108 15208 83 193 11374 15841
84 1333 5056 12441 85 8026 17906 18037 86 162 4432 8739 87 4762 5268 8940 88 3494 6384 17600
89 3029 6710 11442 90 919 9289 19407 91 9130 17762 20598 92 5140 17911 20878 93 14153 16376
19323 94 3863 4847 21567 95 2608 4840 14455 96 1117 4061 5255 97 13468 14536 14928 98 9068
13023 13346 99 1139 14402 15445 100 2190 17004 17906 101 2989 5524 12399 102 8489 8899 15486
103 6683 6970 13387 104 3745 9975 17753 105 1250 4246 17193 106 9992 19441 19965 107 5796
7986 21297 108 2781 3232 16020 109 6891 7654 15969 110 4002 13033 19217 111 7439 9192 13183
112 1390 8673 18485 113 24 4845 14633 114 6083 14765 15640 115 9652 13452 21404 116 7787
10616 17371 117 2959 6783 13581 118 11596 18575 20878 119 8318 14614 20870

(134) As another example of results designed in consideration of these additional conditions, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is 12/15, M is 360, and $Q_{\text{sub.ldpc}}$ is 36, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 16.

(135) TABLE-US-00016 TABLE 16 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0 584 1472 1621 1867 3338 3568 3723 4185 5126 5889 7737 8632 8940 9725 1 221 445 590 3779 3835 6939 7743 8280 8448 8491 9367 10042 11242 12917 2 4662 4837 4900 5029 6449 6687 6751 8684 9936 11681 11811 11886 12089 12909 3 2418 3018 3647 4210 4473 7447 7502 9490 10067 11092 11139 11256 12201 12383 4 2591 2947 3349 3406 4417 4519 5176 6672 8498 8863 9201 11294 11376 12184 5 27 101 197 290 871 1727 3911 5411 6676 8701 9350 10310 10798 12439 6 1765 1897 2923 3584 3901 4048 6963 7054 7132 9165 10184 10824 11278 12669 7 2183 3740 4808 5217 5660 6375 6787 8219 8466 9037 10353 10583 11118 12762 8 73 1594 2146 2715 3501 3572 3639 3725 6959 7187 8406 10120 10507 10691 9 240 732 1215 2185 2788 2830 3499 3881 4197 4991 6425 7061 9756 10491 10 831 1568 1828 3424 4319 4516 4639 6018 9702 10203 10417 11240 11518 12458 11 2024 2970 3048 3638 3676 4152 5284 5779 5926 9426 9945 10873 11787 11837 12 1049 1218 1651 2328 3493 4363 5750 6483 7613 8782 9738 9803 11744 11937 13 1193 2060 2289 2964 3478 4592 4756 6709 7162 8231 8326 11140 11908 12243 14 978 2120 2439 3338 3850 4589 6567 8745 9656 9708 10161 10542 10711 12639 15 2403 2938 3117 3247 3711 5593 5844 5932 7801 10152 10226 11498 12162 12941 16 1781 2229 2276 2533 3582 3951 5279 5774 7930 9824 10920 11038 12340 12440 17 289 384 1980 2230 3464 3873 5958 8656 8942 9006 10175 11425 11745 12530 18 155 354 1090 1330 2002 2236 3559 3705 4922 5958 6576 8564 9972 12760 19 303 876 2059 2142 5244 5330 6644 7576 8614 9598 10410 10718 11033 12957 20 3449 3617 4408 4602 4727 6182 8835 8928 9372 9644 10237 10747 11655 12747 21 811 2565 2820 8677 8974 9632 11069 11548 11839 12107 12411 12695 12812 12890 22 972 4123 4943 6385 6449 7339 7477 8379 9177 9359 10074 11709 12552 12831 23 842 973 1541 2262 2905 5276 6758 7099 7894 8128 8325 8663 8875 10050 24 474 791 968 3902 4924 4965 5085 5908 6109 6329 7931 9038 9401 10568 25 1397 4461 4658 5911 6037 7127 7318 8678 8924 9000 9473 9602 10446 12692 26 1334 7571 12881 27 1393 1447 7972 28 633 1257 10597 29 4843 5102 11056 30 3294 8015 10513 31 1108 10374 10546 32 5353 7824 10111 33 3398 7674 8569 34 7719 9478 10503 35 2997 9418 9581 36 5777 6519 11229 37 1966 5214 9899 38 6 4088 5827 39 836 9248 9612 40 483 7229 7548 41 7865 8289 9804 42 2915 11098 11900 43 6180 7096 9481 44 1431 6786 8924 45 748 6757 8625 46 3312 4475 7204 47 1852 8958 11020 48 1915 2903 4006 49 6776 10886 12531 50 2594 9998 12742 51 159 2002 12079 52 853 3281 3762 53 5201 5798 6413 54 3882 6062 12047 55 4133

6775 9657 56 228 6874 11183 57 7433 10728 10864 58 7735 8073 12734 59 2844 4621 11779 60 3909
7103 12804 61 6002 9704 11060 62 5864 6856 7681 63 3652 5869 7605 64 2546 2657 4461 65 2423
4203 9111 66 244 1855 4691 67 1106 2178 6371 68 391 1617 10126 69 250 9259 10603 70 3435 4614
6924 71 1742 8045 9529 72 7667 8875 11451 73 4023 6108 6911 74 8621 10184 11650 75 6726 10861
12348 76 3228 6302 7388 77 1 1137 5358 78 381 2424 8537 79 3256 7508 10044 80 1980 2219 4569 81
2468 5699 10319 82 2803 3314 12808 83 8578 9642 11533 84 829 4585 7923 85 59 329 5575 86 1067
5709 6867 87 1175 4744 12219 88 109 2518 6756 89 2105 10626 11153 90 5192 10696 10749 91 6260
7641 8233 92 2998 3094 11214 93 3398 6466 11494 94 6574 10448 12160 95 2734 10755 12780 96
1028 7958 10825 97 8545 8602 10793 98 392 3398 11417 99 6639 9291 12571 100 1067 7919 8934 101
1064 2848 12753 102 6076 8656 12690 103 5504 6193 10171 104 1951 7156 7356 105 4389 4780 7889
106 526 4804 9141 107 1238 3648 10464 108 2587 5624 12557 109 5560 5903 11963 110 1134 2570
3297 111 10041 11583 12157 112 1263 9585 12912 113 3744 7898 10646 114 45 9074 10315 115 1051
6188 10038 116 2242 8394 12712 117 3598 9025 12651 118 2295 3540 5610 119 1914 4378 12423 120
1766 3635 12759 121 5177 9586 11143 122 943 3590 11649 123 4864 6905 10454 124 5852 6042
10421 125 6095 8285 12349 126 2070 7171 8563 127 718 12234 12716 128 512 10667 11353 129 3629
6485 7040 130 2880 8865 11466 131 4490 10220 11796 132 5440 8819 9103 133 5262 7543 12411 134
516 7779 10940 135 2515 5843 9202 136 4684 5994 10586 137 573 2270 3324 138 7870 8317 10322
139 6856 7638 12909 140 1583 7669 10781 141 8141 9085 12555 142 3903 5485 9992 143 4467 11998
12904

(136) As another example of results designed in consideration of these additional conditions, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is $6/15$, M is 360, and $Q_{\text{sub.ldpc}}$ is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 17.

(137) TABLE-US-00017 TABLE 17 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0 1606 3402 4961 6751 7132 11516 12300 12482 12592 13342 13764 14123 21576
23946 24533 25376 25667 26836 31799 34173 35462 36153 36740 37085 37152 37468 37658 1 4621
5007 6910 8732 9757 11508 13099 15513 16335 18052 19512 21319 23663 25628 27208 31333 32219
33003 33239 33447 36200 36473 36938 37201 37283 37495 38642 2 16 1094 2020 3080 4194 5098
5631 6877 7889 8237 9804 10067 11017 11366 13136 13354 15379 18934 20199 24522 26172 28666
30386 32714 36390 37015 37162 3 700 897 1708 6017 6490 7372 7825 9546 10398 16605 18561
18745 21625 22137 23693 24340 24966 25015 26995 28586 28895 29687 33938 34520 34858 37056
38297 4 159 2010 2573 3617 4452 4958 5556 5832 6481 8227 9924 10836 14954 15594 16623 18065
19249 22394 22677 23408 23731 24076 24776 27007 28222 30343 38371 5 3118 3545 4768 4992 5227
6732 8170 9397 10522 11508 15536 20218 21921 28599 29445 29758 29968 31014 32027 33685 34378
35867 36323 36728 36870 38335 38623 6 1264 4254 6936 9165 9486 9950 10861 11653 13697 13961
15164 15665 18444 19470 20313 21189 24371 26431 26999 28086 28251 29261 31981 34015 35850
36129 37186 7 111 1307 1628 2041 2524 5358 7988 8191 10322 11905 12919 14127 15515 15711
17061 19024 21195 22902 23727 24401 24608 25111 25228 27338 35398 37794 38196 8 961 3035
7174 7948 13355 13607 14971 18189 18339 18665 18875 19142 20615 21136 21309 21758 23366
24745 25849 25982 27583 30006 31118 32106 36469 36583 37920 9 2990 3549 4273 4808 5707 6021
6509 7456 8240 10044 12262 12660 13085 14750 15680 16049 21587 23997 25803 28343 28693
34393 34860 35490 36021 37737 38296 10 955 4323 5145 6885 8123 9730 11840 12216 19194 20313
23056 24248 24830 25268 26617 26801 28557 29753 30745 31450 31973 32839 33025 33296 35710
37366 37509 11 264 605 4181 4483 5156 7238 8863 10939 11251 12964 16254 17511 20017 22395
22818 23261 23422 24064 26329 27723 28186 30434 31956 33971 34372 36764 38123 12 520 2562
2794 3528 3860 4402 5676 6963 8655 9018 9783 11933 16336 17193 17320 19035 20606 23579 23769
24123 24966 27866 32457 34011 34499 36620 37526 13 10106 10637 10906 34242 14 1856 15100
19378 21848 15 943 11191 27806 29411 16 4575 6359 13629 19383 17 4476 4953 18782 24313 18
5441 6381 21840 35943 19 9638 9763 12546 30120 20 9587 10626 11047 25700 21 4088 15298 28768
35047 22 2332 6363 8782 28863 23 4625 4933 28298 30289 24 3541 4918 18257 31746 25 1221 25233
26757 34892 26 8150 16677 27934 30021 27 8500 25016 33043 38070 28 7374 10207 16189 35811 29
611 18480 20064 38261 30 25416 27352 36089 38469 31 1667 17614 25839 32776 32 4118 12481
21912 37945 33 5573 13222 23619 31271 34 18271 26251 27182 30587 35 14690 26430 26799 34355

36 13688 16040 20716 34558 37 2740 14957 23436 32540 38 3491 14365 14681 36858 39 4796 6238
25203 27854 40 1731 12816 17344 26025 41 19182 21662 23742 27872 42 6502 13641 17509 34713
43 12246 12372 16746 27452 44 1589 21528 30621 34003 45 12328 20515 30651 31432 46 3415
22656 23427 36395 47 632 5209 25958 31085 48 619 3690 19648 37778 49 9528 13581 26965 36447
50 2147 26249 26968 28776 51 15698 18209 30683 52 1132 19888 34111 53 4608 25513 38874 54 475
1729 34100 55 7348 32277 38587 56 182 16473 33082 57 3865 9678 21265 58 4447 20151 27618 59
6335 14371 38711 60 704 9695 28858 61 4856 9757 30546 62 1993 19361 30732 63 756 28000 29138
64 3821 24076 31813 65 4611 12326 32291 66 7628 21515 34995 67 1246 13294 30068 68 6466 33233
35865 69 14484 23274 38150 70 21269 36411 37450 71 23129 26195 37653

(138) As another example of results designed in consideration of these additional conditions, if a codeword length $N_{\text{sub.ldpc}}$ is 64800, a code rate R is 10/15, M is 360, and $Q_{\text{sub.ldpc}}$ is 60, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 18.

(139) TABLE-US-00018 TABLE 18 i Indices of rows at which 1s are positioned in 0-th column of i -th column group 0 979 1423 4166 4609 6341 8258 10334 10548 14098 14514 17051 17333 17653 17830 17990 1 2559 4025 6344 6510 9167 9728 11312 14856 17104 17721 18600 18791 19079 19697 19840 2 3243 6894 7950 10539 12042 13233 13938 14752 16449 16727 17025 18297 18796 19400 21577 3 3272 3574 6341 6722 9191 10807 10957 12531 14036 15580 16651 17007 17309 19415 19845 4 155 4598 10201 10975 11086 11296 12713 15364 15978 16395 17542 18164 18451 18612 20617 5 1128 1999 3926 4069 5558 6085 6337 8386 10693 12450 15438 16223 16370 17308 18634 6 2408 2929 3630 4357 5852 7329 8536 8695 10603 11003 14304 14937 15767 18402 21502 7 199 3066 6446 6849 8973 9536 10452 12857 13675 15913 16717 17654 19802 20115 21579 8 312 870 2095 2586 5517 6196 6757 7311 7368 13046 15384 18576 20349 21424 21587 9 985 1591 3248 3509 3706 3847 6174 6276 7864 9033 13618 15675 16446 18355 18843 10 975 3774 4083 5825 6166 7218 7633 9657 10103 13052 14240 17320 18126 19544 20208 11 1795 2005 2544 3418 6148 8051 9066 9725 10676 10752 11512 15171 17523 20481 21059 12 167 315 1824 2325 2640 2868 6070 6597 7016 8109 9815 11608 16142 17912 19625 13 1298 1896 3039 4303 4690 8787 12241 13600 14478 15492 16602 17115 17913 19466 20597 14 568 3695 6045 6624 8131 8404 8590 9059 9246 11570 14336 18657 18941 19218 21506 15 228 1889 1967 2299 3011 5074 7044 7596 7689 9534 10244 10697 11691 17902 21410 16 1330 1579 1739 2234 3701 3865 5713 6677 7263 11172 12143 12765 17121 20011 21436 17 303 1668 2501 4925 5778 5985 9635 10140 10820 11779 11849 12058 15650 20426 20527 18 698 2484 3071 3219 4054 4125 5663 5939 6928 7086 8054 12173 16280 17945 19302 19 232 1619 3040 4901 7438 8135 9117 9233 10131 13321 17347 17436 18193 18586 19929 20 12 3721 6254 6609 7880 8139 10437 12262 13928 14065 14149 15032 15694 16264 18883 21 482 915 1548 1637 6687 9338 10163 11768 11970 15524 15695 17386 18787 19210 19340 22 1291 2500 4109 4511 5099 5194 10014 13165 13256 13972 15409 16113 16214 18584 20998 23 1761 4778 7444 7740 8129 8341 8931 9136 9207 10003 10678 13959 17673 18194 20990 24 3060 3522 5361 5692 6833 8342 8792 11023 11211 11548 11914 13987 15442 15541 19707 25 1322 2348 2970 5632 6349 7577 8782 9113 9267 9376 12042 12943 16680 16970 21321 26 6785 11960 21455 27 1223 15672 19550 28 5976 11335 20385 29 2818 9387 15317 30 2763 3554 18102 31 5230 11489 18997 32 5809 15779 20674 33 2620 17838 18533 34 3025 9342 9931 35 3728 5337 12142 36 2520 6666 9164 37 12892 15307 20912 38 10736 12393 16539 39 1075 2407 12853 40 4921 5411 18206 41 5955 15647 16838 42 6384 10336 19266 43 429 10421 17266 44 4880 10431 12208 45 2910 11895 12442 46 7366 18362 18772 47 4341 7903 14994 48 4564 6714 7378 49 4639 8652 18871 50 15787 18048 20246 51 3241 11079 13640 52 1559 2936 15881 53 2737 6349 10881 54 10394 16107 17073 55 8207 9043 12874 56 7805 16058 17905 57 11189 15767 17764 58 5823 12923 14316 59 11080 20390 20924 60 568 8263 17411 61 1845 3557 6562 62 2890 10936 14756 63 9031 14220 21517 64 3529 12955 15902 65 413 6750 8735 66 6784 12092 16421 67 12019 13794 15308 68 12588 15378 17676 69 8067 14589 19304 70 1244 5877 6085 71 15897 19349 19993 72 1426 2394 12264 73 3456 8931 12075 74 13342 15273 20351 75 9138 13352 20798 76 7031 7626 14081 77 4280 4507 15617 78 4170 10569 14335 79 3839 7514 16578 80 4688 12815 18782 81 4861 7858 9435 82 605 5445 12912 83 2280 4734 7311 84 6668 8128 12638 85 3733 10621 19534 86 13933 18316 19341 87 1786 3037 21566 88 2202 13239 16432 89 4882 5808 9300 90 4580 8484 16754 91 14630 17502 18269 92 6889 11119 12447 93 8162 9078 16330 94 6538 17851 18100 95 17763 19793

20816 96 21807 17567 97 6640 14428 15175 98 877 12035 14081 99 1336 6468 12328 100 5948
9146 12003 101 3782 5699 12445 102 1770 7946 8244 103 7384 12639 14989 104 1469 11586 20959
105 7943 10450 15907 106 5005 8153 10035 107 17750 18826 21513 108 4725 8041 10112 109 3837
16266 17376 110 11340 17361 17512 111 1269 4611 4774 112 2322 10813 16157 113 16752 16843
18959 114 70 4325 18753 115 3165 8153 15384 116 160 8045 16823 117 14112 16724 16792 118 4291
7667 18176 119 5943 19879 20721

(140) Here, it is to be noted that even if the parity-check matrix in which an order of indices within a sequence for each i-th column group in any of above Tables 9 to 18 is changed, the changed parity-check matrix may also be applied to the same codes to which the original parity-check matrix applies.

(141) For example, as shown in above Table 11, the sequence corresponding to the zero (0)-th column group is arranged in an order of 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140, and even if the sequence has a changed index order like 2135, 8943, 4635, 114, 3045, 10109, 13161, 21140, 5681, 6571, 5512, 19328, 14218, 13668, 17417, 10053, a parity-check matrix having the changed sequence may be used for the same codes.

(142) In addition, even if one index sequence of one column group is changed to an index sequence of another column group and vice versa, that is, an index sequence of the other column group is changed to the index sequence of the first column group, in above Tables 9 to 18, algebraic characteristics such as the cycle characteristics and the degree distributions on a graph of codes are not changed. Therefore, this case of changing index sequences between column groups of a parity-check matrix is another exemplary embodiment of the inventive concept.

(143) For example, in above Table 11, the index sequence 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140 of the zero (0)-th column group and the index sequence 19, 768, 1263, 3305, 6513, 7677, 7956, 9040, 13427, 16641, 17280, 18452, 18584, 18925, 19559, 20587 of a twelfth column group can be changed to set the sequence of the zero (0)-th column group to be 19, 768, 1263, 3305, 6513, 7677, 7956, 9040, 13427, 16641, 17280, 18452, 18584, 18925, 19559, 20587, and the sequence of the twelfth column group is set to be 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140. That is, even if the index sequences are exchanged between the zero (0)-th column group and the twelfth column group, the cycle characteristics, the degree distributions, and the like in terms of the graph of codes are not changed (actually, since exchanging the index sequences between corresponding column groups is the same as changing only the arranging order of column groups within a parity-check matrix, the main algebraic characteristics are not changed).

(144) In addition, a parity-check matrix in which an integer multiple of $Q_{\text{sub}} \cdot \text{ldpc}$ is added to all indices of any column group in above Tables 9 to 18 may also result in the same algebraic characteristics such as the cycle characteristics and the degree distributions on the graph of the codes, according to an exemplary embodiment.

(145) For example, if a multiple of 60 (that is, $Q_{\text{sub}} \cdot \text{ldpc} = (N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}) / M = 60$) is added to all indices 12575, 15845 and 18200 of a twenty-fourth column group in above Table 11, that is, the indices are changed to 12635 ($=12575+60$), 15905 ($=15845+60$), and 18260 ($=18200+60$), the algebraic characteristics such as the cycle characteristics and the degree distributions on the graph of codes are not changed (actually, since a sequence obtained by adding an integer multiple of $Q_{\text{sub}} \cdot \text{ldpc}$ in a column group has the same effect as rearranging only the order of columns within the column group, the main algebraic characteristics are not changed).

(146) Here, it is to be noted that in the case in which an index value obtained by adding an integer multiple of $Q_{\text{sub}} \cdot \text{ldpc}$ to a given index sequence is a value of $N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}$ or more, the index value is changed and applied to a value obtained by performing a modulo operation on $N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc}$.

(147) For example, since a result of adding only 60×60 to the sequence 12575, 15845 and 18200 of the twenty-fourth column group in above Table 11 becomes 16175, 19445 and 21800 while $N_{\text{sub}} \cdot \text{ldpc} - K_{\text{sub}} \cdot \text{ldpc} = 21600$, the changed sequence may be 16175, 19445 and 200 or 200, 16175 and 19445 by applying modulo-21600 to a result of adding only the integer multiple of $Q_{\text{sub}} \cdot \text{ldpc}$ to the given sequence.

(148) Hereinafter, a process of encoding LDPC codes using a parity-check matrix having the structure as

illustrated in FIG. 3 will be described. As described above, the process of encoding LDPC codes is to determine a codeword C meeting a relational equation: $\text{parity-check matrix} \times \text{codeword} = 0$. That is, the process of encoding LDPC codes may be represented by $H \cdot \text{Math.C} \cdot \text{sup.T} = 0$. Here, H is the parity-check matrix and C represents the LDPC codeword.

(149) Hereinafter, if it is assumed that LDPC encoded information word bits are $(i_{\text{sub}.0}, i_{\text{sub}.1}, \dots, i_{\text{sub}.K_{\text{sub}}.\text{ldpc.sub.}-1})$ and the LDPC codeword bits generated by the LDPC encoding are $(c_{\text{sub}.0}, c_{\text{sub}.1}, \dots, c_{\text{sub}.N_{\text{sub}}.\text{ldpc.sub.}-1})$, a method for calculating LDPC codeword bits will be described.

(150) First, since the LDPC code is a systematic code, $c_{\text{sub}.k}$ for $0 < k < K_{\text{sub}}.\text{ldpc}-1$ is set to be the same as $i_{\text{sub}.k}$. In addition, the remaining codeword bits are set to be $p_{\text{sub}.k} := c_{\text{sub}.k + K_{\text{sub}}.\text{ldpc}}$. Here, $p_{\text{sub}.k}$ is parity bits and may be calculated as described below.

(151) Meanwhile, according to an exemplary embodiment, since the parity-check matrix is defined by in above Tables 9 to 18, a process to be described below may be applied in the case in which the parity-check matrix is defined by in above Tables 9 to 18.

(152) First, if it is assumed that an entry notated in a j -th position of an i -th row in above Tables 9 to 18 is $q(i, j, l) = q(i, j, 0) + Q_{\text{sub}.\text{ldpc}} \cdot \text{Math.1}(\text{mod } N_{\text{sub}.\text{ldpc}} - K_{\text{sub}.\text{ldpc}})$ for $0 < l < 360$. Here, accumulation '+' means additions defined in a Galois field (GF) (2) (that is, additions in GF (2)). In addition, $Q_{\text{sub}.\text{ldpc}}$, which is a size of each column cyclically shifted in an information word sub-matrix, may be a value defined in above Tables 9 to 18, respectively.

(153) Meanwhile, when $q(i, j, 0)$ and $q(i, j, l)$ are defined as described above, a process of calculating parity bits is as follows. Step 1) The parity bits are initialized to '0'. That is, $p_{\text{sub}.k} = 0$ for $0 < k < N_{\text{sub}.\text{ldpc}} - K_{\text{sub}.\text{ldpc}}$. Step 2) i and l are set to be $i: \lfloor k/360 \rfloor$ and $l: k \pmod{360}$ for all k values of $0 < k < K_{\text{sub}.\text{ldpc}}$. Here, $\lfloor x \rfloor$ is the largest integer value among integers that are not larger than x . That is, $\lfloor 1.2 \rfloor = 1$. Next, $i_{\text{sub}.k}$ is added to $p_{\text{sub}.q(i, j, l)}$ for all j s as following based on the set i and l values. That is, $p_{\text{sub}.q(i, 0, l)} = p_{\text{sub}.q(i, 0, l)} + i_{\text{sub}.k}$, $p_{\text{sub}.q(i, 1, l)} = p_{\text{sub}.q(i, 1, l)} + i_{\text{sub}.k}$, $p_{\text{sub}.q(i, 2, l)} = p_{\text{sub}.q(i, 2, l)} + i_{\text{sub}.k}$, $\dots$, $p_{\text{sub}.q(i, w(i)-1, l)} = p_{\text{sub}.q(i, w(i)-1, l)} + i_{\text{sub}.k}$ are calculated.

(154) Here, $w(i)$ is the number of values of the i -th row in above Tables 9 to 18 and represents the number of ones (1s) in a column corresponding to $i_{\text{sub}.k}$ in the parity-check matrix. In addition, $q(i, j, 0)$, which is the entry notated in the j -th position of the i -th row in above Tables 9 to 18, is an index of the parity bit and represents a position of a row at which one (1) is positioned in a column corresponding to $i_{\text{sub}.k}$ in the parity-check matrix. Step 3) $p_{\text{sub}.k} = p_{\text{sub}.k} + p_{\text{sub}.k-1}$ is calculated for all k s meeting $0 < k < N_{\text{sub}.\text{ldpc}} - K_{\text{sub}.\text{ldpc}}$ to calculate the parity bits $p_{\text{sub}.k}$.

(155) The parity bits are calculated by the foregoing method, such that all the LDPC codeword bits $c_{\text{sub}.0}, c_{\text{sub}.1}, \dots, c_{\text{sub}.N_{\text{sub}}.\text{ldpc.sub.}-1}$ may be calculated.

(156) Meanwhile, the LDPC encoding process as described above is only an example. That is, since the LDPC encoding process is a process of calculating an LDPC codeword C satisfying $H \cdot \text{Math.C} \cdot \text{sup.T} = 0$, various encoding methods for the given parity-check matrix may be present.

(157) For example, a scheme applied in the DVB-T2 standard may also be applied to the case in which the parity-check matrix is defined by above Tables 9 to 18. Hereinafter, the LDPC encoding process according to the scheme described in the DVB-T2 standard will be schematically described using an example in which the parity-check matrix is defined in above Table 11.

(158) First, if it is assumed that information word bits having a length of $K_{\text{sub}.\text{ldpc}}$ are $[i_{\text{sub}.0}, i_{\text{sub}.1}, i_{\text{sub}.2}, \dots, i_{\text{sub}.K_{\text{sub}}.\text{ldpc.sub.}-1}]$ and parity bits having a length of $N_{\text{sub}.\text{ldpc}} - K_{\text{sub}.\text{ldpc}}$ are $[p_{\text{sub}.0}, p_{\text{sub}.1}, p_{\text{sub}.2}, \dots, p_{\text{sub}.N_{\text{sub}}.\text{ldpc.sub.}-K_{\text{sub}}.\text{ldpc.sub.}-1}]$, the LDPC encoding may be performed by the following process. Step 1) The parity bits are initialized to '0'. That is, $p_{\text{sub}.0} = p_{\text{sub}.1} = p_{\text{sub}.2} = \dots = p_{\text{sub}.N_{\text{sub}}.\text{ldpc.sub.}-K_{\text{sub}}.\text{ldpc.sub.}-1}$. Step 2) A zero (0)-th information word bit $i_{\text{sub}.0}$ is accumulated in a parity bit which has an address of a parity bit defined in a first row (that is, a row of $i=0$) of above Table 11 as the index of the parity bit. This may be represented by following mathematical expressions 11.

$$\begin{aligned}
(159) \quad & p_{114} = p_{114} \oplus i_0 & p_{10053} = p_{10053} \oplus i_0 \\
& p_{2135} = p_{2135} \oplus i_0 & p_{10109} = p_{10109} \oplus i_0 \\
& p_{3045} = p_{3045} \oplus i_0 & p_{13161} = p_{13161} \oplus i_0 \\
& p_{4635} = p_{4635} \oplus i_0 & p_{13668} = p_{13668} \oplus i_0 \\
& p_{5512} = p_{5512} \oplus i_0 & p_{14218} = p_{14218} \oplus i_0 \\
& p_{5681} = p_{5681} \oplus i_0 & p_{17417} = p_{17417} \oplus i_0 \\
& p_{6571} = p_{6571} \oplus i_0 & p_{19328} = p_{19328} \oplus i_0 \\
& p_{8943} = p_{8943} \oplus i_0 & p_{21140} = p_{21140} \oplus i_0
\end{aligned} \tag{11}$$

(160) In the above mathematical expressions, $i_{\text{sub}.0}$ represents a zero (0)-th information word bit, $p_{\text{sub}.i}$ represents an i -th parity bit, and $\oplus$ represents a binary operation. According to the binary operation, $\text{sub}.1 \oplus \text{sub}.1$ is $\text{sub}.1 \oplus \text{sub}.0$ is 1, $\text{sub}.0 \oplus \text{sub}.1$ is 1, and $\text{sub}.0 \oplus \text{sub}.0$ is 0. Step 3) The remaining 359 information word bits $i_{\text{sub}.m}$ ($m=1, 2, \dots, 359$) are accumulated in the parity bits. Here, the remaining information word bits may be information word bits which belong to the same column group as a column group to which $i_{\text{sup}.0}$ belongs. In this case, the addresses of the parity bits may be determined based on following mathematical expression 12.

$$(x + (m \bmod 360) \times Q_{\text{sub}.ldpc}) \bmod (N_{\text{sub}.ldpc} - K_{\text{sub}.ldpc}) \tag{12}$$

(161) In above mathematical expression 12, x is an address of a parity bit accumulator corresponding to the information word bit $i_{\text{sub}.0}$, and $Q_{\text{sub}.ldpc}$ is 60 as a size of each column which is shifted in a sub-matrix corresponding to the information word.

(162) As a result, each of the information word bits $i_{\text{sub}.m}$ ($m=1, 2, \dots, 359$) is accumulated in each parity bit having the addresses of the parity bits calculated based on above mathematical expression 12 as indices. As an example, operations as represented by following mathematical expression 13s may be performed on the information word bit

$$\begin{aligned}
(163) \quad & p_{174} = p_{174} \oplus i_1 & p_{10113} = p_{10113} \oplus i_1 \\
& p_{2195} = p_{2195} \oplus i_1 & p_{10169} = p_{10169} \oplus i_1 \\
& p_{3105} = p_{3105} \oplus i_1 & p_{13221} = p_{13221} \oplus i_1 \\
& p_{4695} = p_{4695} \oplus i_1 & p_{13728} = p_{13728} \oplus i_1 \\
& p_{5572} = p_{5572} \oplus i_1 & p_{14278} = p_{14278} \oplus i_1 \\
& p_{5741} = p_{5741} \oplus i_1 & p_{17477} = p_{17477} \oplus i_1 \\
& p_{6631} = p_{6631} \oplus i_1 & p_{19388} = p_{19388} \oplus i_1 \\
& p_{9003} = p_{9003} \oplus i_1 & p_{21200} = p_{21200} \oplus i_1
\end{aligned} \tag{13}$$

(164) In above mathematical expression, $i_{\text{sub}.1}$ represents a 1-th information word bit, $p_{\text{sub}.i}$ represents an i -th parity bit, and $\oplus$ represents a binary operation. According to the binary operation, $\text{sub}.1 \oplus \text{sub}.1$ is 0, $\text{sub}.1 \oplus \text{sub}.0$ is 1, $\text{sub}.0 \oplus \text{sub}.1$ is 1, and $\text{sub}.0 \oplus \text{sub}.0$ is 0. Step 4) A 360-th information word bit $i_{\text{sub}.360}$ is accumulated in a parity bit which has an address of a parity bit defined in a second row (that is, a row of $i=1$) of above Table 11 as the index of the parity bit. Step 5) The remaining 359 information word bits belonging to the same group as a group to which the information word bit $i_{\text{sub}.360}$ belongs are accumulated in the parity bit. In this case, the address of the parity bit may be determined based on above mathematical expression 6. However, in this case, x is an address of a parity bit accumulator corresponding to the information word bit $i_{\text{sub}.360}$. Step 6) the foregoing processes of Step 4 and Step 5 are repeated for all of the column groups of above Table 5. Step 7) As a result, the parity bit $p_{\text{sub}.i}$ is calculated based on following mathematical expression 14. Here, i is initialized to one (1).

$$p_{\text{sub}.i} = p_{\text{sub}.i} \oplus p_{\text{sub}.i-1} \quad i=1, 2, \dots, N_{\text{sub}.ldpc} - K_{\text{sub}.ldpc} - 1 \tag{14}$$

(165) In above mathematical expression 14, $p_{\text{sub}.i}$ represents an i -th parity bit, $N_{\text{sub}.ldpc}$ represents the LDPC codeword length, $K_{\text{sub}.ldpc}$ represents the information word length in the LDPC codeword, and

⊕ represents a binary operation. According to the binary operation, $.sub.1 \oplus .sub.1$ is 0, $.sub.1 \oplus .sub.0$ is 1, $.sub.0 \oplus .sub.1$ is 1, and $.sub.0 \oplus .sub.0$ is 0.

(166) According to the above method, the parity bits may be calculated.

(167) Meanwhile, the addresses of the parity bit present in the zero (0)-th column of the i-th column group are the same as the indices of the row at which one (1) is positioned in the zero (0)-th column of the i-th column group. Therefore, the indices of the row at which one (1) is positioned in the zero (0)-th column of the i-th column group in above Tables 9 to 18 are represented as addresses of the parity bits in the encoding process. Therefore, above Tables 9 to 18 may show “addresses of parity bit accumulators”.

(168) As described above, according to the above exemplary embodiments, the LDPC encoding process may be performed using various schemes to generate an LDPC codeword.

(169) Meanwhile, LDPC codes may be decoded using an iterative decoding algorithm based on the sum-product algorithm on the bipartite graph illustrated in FIG. 2, in which the sum-product algorithm is a kind of message passing algorithm.

(170) Hereinafter, a message passing operation generally used for LDPC decoding will be described with reference to FIGS. 5A and 5B.

(171) FIGS. 5A and 5B illustrate a message passing operation in any check node and variable node for LDPC decoding, according to an exemplary embodiment.

(172) FIG. 5A illustrates a check node m 500 and a plurality of variable nodes 510, 520, 530 and 540 connected to the check node m 500. In addition, $T_{n',m}$ illustrated in FIG. 5A indicates a message passing from the variable node n' 510 to the check node m 500, and $E_{n,m}$ indicates a message passing from the check node m 500 to the variable node n 530. Here, a set of all variable nodes connected to the check node m 500 is defined by $N(m)$, and a set except the variable node n 530 in the $N(m)$ is defined by $N(m) \setminus n$.

(173) In this case, a message update rule based on the sum-product algorithm may be represented by following mathematical expressions 15.

$$(174) \quad E_{n,m} = \left[\prod_{n' \in N(m) \setminus n} T_{n',m} \right] \text{Sign}(E_{n,m}) = \prod_{n' \in N(m) \setminus n} \text{sign}(T_{n',m}) \quad (15)$$

(175) In above mathematical expressions 15, $\text{Sign}(E_{n,m})$ indicates a sign of the message $E_{n,m}$, and $E_{n,m}$ indicates a magnitude of the message $E_{n,m}$. Meanwhile, a function $\Phi(x)$ may be represented by following mathematical expression 16.

$$(176) \quad \Phi(x) = -\log(\tanh(\frac{x}{2})) \quad (16)$$

(177) FIG. 5B illustrates a variable node x 550 and a plurality of check nodes 560, 570, 580 and 590 connected to the variable node x 550. In addition, $E_{y',x}$ illustrated in FIG. 5B indicates a message passing from the check node y' 560 to the variable node x 550, and $T_{y,x}$ indicates a message passing from the variable node x 550 to the check node y 580. Here, a set of all check nodes connected to the variable node x 550 is defined by $M(x)$, and a set except the check node y 580 in $M(x)$ is defined by $M(x) \setminus y$. In this case, the message update rule based on the sum-product algorithm may be represented by following mathematical expression 17.

$$T_{y,x} = E_{y',x} + \sum_{y' \in M(x) \setminus y} E_{y',x} \quad (17)$$

(178) In above mathematical expression 17, $E_{y',x}$ represents an initial message value of the variable node x.

(179) In addition, when a bit value of the node x is decided, it may be represented by following mathematical expression 18.

$$P_x = E_x + \sum_{y' \in M(x)} E_{y',x} \quad (18)$$

(180) In this case, an encoding bit corresponding to the node x may be decided depending on a value of P_x .

(181) In FIG. 5, the foregoing method is a general decoding method, and thus, the detailed description thereof will be omitted. However, in addition to the method described with reference to FIG. 5, other methods for determining the passing message values at the variable node and the check node may also be applied (Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger, “Factor Graphs and the Sum-Product Algorithm,” IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. 47, NO. 2,

(182) FIG. 6 is a block diagram illustrating a configuration of an encoding apparatus according to an exemplary embodiment. The encoding apparatus **600** may perform the LDPC encoding described above.
 (183) As illustrated in FIG. 6, the encoding apparatus **600** includes an LDPC encoder **610**. The LDPC encoder **610** may perform the LDPC encoding on input bits based on the parity-check matrix generated in the above method to generate an LDPC codeword. The LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

(184) Here, the parity-check matrix may have the same structure as that of the parity-check matrix **300** illustrated in FIG. 3.

(185) In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix described above.

(186) Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

(187) In this case, the LDPC encoder **610** may perform the LDPC encoding using parity-check matrices differently defined depending on a code rate (that is, a code rate of the LDPC codes).

(188) For example, the LDPC encoder **610** may perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15, and perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC encoder **610** may perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15, and perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

(189) Meanwhile, a detailed method for performing the LDPC encoding is already described, and thus, duplicate descriptions thereof will be omitted.

(190) The encoding apparatus **600** may further include a memory (not illustrated) in which information on a code rate, a codeword length and a parity-check matrix of an LDPC code is previously stored, and the LDPC encoder **610** may perform LDPC encoding using this information. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

(191) FIG. 7 is a block diagram showing a configuration of a transmitting apparatus according to an exemplary embodiment. As illustrated in FIG. 7, a transmitting apparatus **700** may include a Bose, Chaudhuri, Hocquenghem (BCH) encoder **710**, an LDPC encoder **720**, an interleaver **730**, and a modulator **740**.

(192) The BCH encoder **710** performs BCH encoding on input bits and outputs a BCH codeword generated by BCH encoding to the LDPC encoder **720**.

(193) In detail, the BCH encoder **710** performs the BCH encoding on the input bits $L=[i.sub.0, i.sub.1, \dots, i.sub.K.sub.bch.sub.-1]$, to generate $N.sub.ldpc-K.sub.bch$ BCH parity bits and generate an BCH codeword $l.sub.ldpc=[i.sub.0, i.sub.1, i.sub.2, \dots, i.sub.K.sub.ldpc.sub.-1]$. The BCH codeword $l.sub.ldpc=[i.sub.0, i.sub.1, i.sub.2, \dots, i.sub.K.sub.ldpc.sub.-1]$ is an information word for LDPC encoding, to the LDPC encoder **720**.

(194) The BCH encoding is a well-known technology and is disclosed in "Bose, R. C.; Ray-Chaudhuri, D. K. (March 1960), "On A Class of Error Correcting Binary Group Codes", Information and Control 3 (1): 68-79, ISSN 0890-5401", etc., and thus, detailed descriptions thereof will be omitted herein.

(195) Meanwhile, it may be changed whether the BCH encoder **710** is used. That is, in some cases, the BCH encoder **710** may also be omitted.

(196) The LDPC encoder **720** performs LDPC encoding on the BCH codeword output from the BCH encoder **710** and outputs an LDPC codeword generated by the LDPC encoding to the interleaver **730**.

(197) In detail, the LDPC encoder **720** performs the LDPC encoding on the BCH codeword $l.sub.ldpc=[i.sub.0, i.sub.1, i.sub.2, \dots, i.sub.K.sub.ldpc.sub.-1]$ output from the BCH encoder **710** as the information word to generate $N.sub.ldpc-K.sub.ldpc$ LDPC parity bits and generate the LDPC codeword

c.sub.ldpc=[c.sub.1, . . . ,c.sub.N.sub.ldpc.sub.-1].

(198) However, when the BCH encoder **710** is omitted, the LDPC encoder **720** may perform the LDPC encoding on the input bits.

(199) Meanwhile, the LDPC encoder **720** of FIG. 7 may be implemented as the LDPC encoder **610** described with reference to FIG. 6. That is, the LDPC encoder **720** may perform the LDPC encoding using the parity-check matrix in which the information word sub-matrix is defined by above Tables 9 to 18 depending on the code rate and the parity sub-matrix has the dual diagonal structure.

(200) To this end, the transmitting apparatus **700** may include a memory (not illustrated) for storing the information on the parity-check matrix. In this case, the parity-check matrix may be various depending on the code rate and may be the table defined by above Tables 9 to 18 as an example. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

(201) The interleaver **730** performs interleaving on the LDPC codeword output from the LDPC encoder **720** and outputs the interleaved bits to the modulator **740**.

(202) In this case, the interleaver **730** receives an LDPC codeword bit string output from the LDPC encoder **720** to perform the interleaving using a predetermined scheme. The interleaving scheme may be variously present and it may be variable whether or not the interleaver **730** is used.

(203) The modulator **740** modulates the bit string output from the interleaver **730** and transmits the modulated bit string to a receiving apparatus (for example, **1000** of FIG. 10).

(204) In detail, the modulator **740** may demultiplex bits output from the interleaver **730** and map the demultiplexed bits to constellation.

(205) That is, the modulator **740** may convert the bits output from the interleaver **730** in a serial-to-parallel scheme to generate cells formed of a predetermined number of bits. Here, the number of bits forming each cell may be equal to the number of bits forming modulation symbols which are mapped to the constellation.

(206) Next, the modulator **740** may map the demultiplexed bits to the constellation. That is, the modulator **740** modulates the demultiplexed bits using various modulation schemes such as QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, and 4096-QAM to generate the modulation symbols, and map the generated modulation symbols to constellation points. In this case, the demultiplexed bits are formed of the cells corresponding to the modulation symbols, and thus, each cell may sequentially be mapped to the constellation points.

(207) Further, the modulator **740** may modulate signals mapped to the constellation and transmit the modulated signals to the receiving apparatus **1000**. For example, the modulator **740** may map the signals, which are mapped to the constellation, to an OFDM frame by using an orthogonal frequency division multiplexing (OFDM) scheme and transmit the mapped signals to the receiving apparatus **1000** through an allocated channel.

(208) The transmitting apparatus **700** may pre-store various parameters which are used for encoding, interleaving, and modulation. Here, the parameters used for encoding may be information on the code rate and the codeword length of the BCH codes and the information on the code rate, the codeword length, and the parity-check matrix of the LDPC codes. Further, the parameters used for interleaving may be information on an interleaving rule and the parameter used for modulation may be information on the modulation schemes. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

(209) In this case, each component configuring the transmitting apparatus **700** may be operated using the parameters.

(210) Meanwhile, although not illustrated, the transmitting apparatus **700** may further include a controller (not illustrated) for controlling an operation of the transmitting apparatus **700**.

(211) In this case, the controller (not illustrated) may provide the information on the code rate and the codeword length of the BCH codes to the BCH encoder **710** and provide the information on the code rate, the codeword length, the parity-check matrix of the LDPC codes to the LDPC encoder **720**. Further, the controller (not illustrated) may provide the information on the interleaving schemes to the interleaver

730 and the information on the modulation schemes to the modulator **740**. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

(212) FIG. **8** is a block diagram illustrating a configuration of a decoding apparatus according to an exemplary embodiment. As illustrated in FIG. **8**, a decoding apparatus **800** includes an LDPC decoder **810**.

(213) The LDPC decoder **810** performs LDPC decoding on an LDPC codeword based on a parity-check matrix. Here, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

(214) For example, the LDPC decoder **810** passes log likelihood ratio (LLR) values corresponding to the LDPC codeword bits using an iterative decoding algorithm to perform the LDPC decoding, thereby generating information word bits.

(215) Here, the LLR values may be represented by channel values corresponding to the LDPC codeword bits by various methods.

(216) For example, the LLR values may be represented by values obtained by taking a log on a ratio of a probability that bits transmitted through the channel at the transmitting side are zero (0) and a probability that bits are one (1). Further, the LLR values may be bit values determined by a soft decision and may also be a representative value determined depending on a section to which the probability that the bits transmitted from the transmitting side are zero (0) or one (1) belongs.

(217) In this case, the transmitting side may use the LDPC encoder **610** as illustrated in FIG. **6** to generate the LDPC codeword.

(218) Meanwhile, the parity-check matrix used for the LDPC decoding may have the same form as that of the parity-check matrix **300** illustrated in FIG. **3**.

(219) In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix.

(220) Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and is defined by the table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

(221) In this case, the LDPC decoder **810** may perform the LDPC decoding using the parity-check matrices differently defined depending on a code rate (that is, a code rate of LDPC codes).

(222) For example, the LDPC decoder **810** may perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15, and perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC decoder **810** may perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

(223) As described above, the LDPC decoder **810** may perform the LDPC decoding using an iterative decoding algorithm. In this case, the LDPC decoder **810** may be configured as illustrated in FIG. **9**. However, the iterative decoding algorithm is already known, and thus, the detailed configuration illustrated in FIG. **9** is only one example.

(224) As illustrated in FIG. **9**, a decoding apparatus **900** includes an input processor **911**, a memory **912**, a variable node operator **913**, a controller **914**, a check node operator **915** and an output processor **916**.

(225) The input processor **911** stores an input value. In detail, the input processor **911** may store the LLR values of signals received through a wireless channel.

(226) The controller **914** determines the number of values input to the variable node operator **913**, an address value in the memory **912**, the number of values input to the check node operator **915**, an address value in the memory **912**, and the like, based on a size (that is, a codeword length) of a block of the signal received through the wireless channel and a parity-check matrix corresponding to the code rate.

(227) According to the exemplary embodiment, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group may perform the decoding based on the parity-check matrices

defined by above Tables 9 to 18.

(228) The memory **912** stores the input data and output data of the variable node operator **913** and the check node operator **915**.

(229) The variable node operator **913** receives data from the memory **912** based on information on the address of the input data and information on the number of input data which are received from the controller **914** to perform a variable node operation. Next, the variable node operator **913** stores variable node operation results in the memory **912** based on information on the address of the output data and information on the number of output data which are received from the controller **914**. Further, the variable node operator **913** inputs the variable node operation results to the output processor **916** based on data which are received from the input processor **911** and the memory **912**. Here, the variable node operation is already described with reference to FIG. 5.

(230) The check node operator **915** receives data from the memory **912** based on the information on the address of the input data and the information on the number of input data which are received from the controller **914** to perform a variable node operation. Next, the check node operator **915** stores variable node operation results in the memory **912** based on information on the address of the output data and information on the number of output data which are received from the controller **914**. Here, the check node operation is already described with reference to FIG. 5.

(231) The output processor **916** performs a soft decision on whether information word bits of the codeword at the transmitting side are zero (0) or one (1) based on data received from the variable node operator **913** and then outputs soft-decision results, such that an output value of the output processor **916** may be a value which is finally decoded. In this case, in FIG. 5, the soft decision may be performed based on a value obtained by adding all message values (an initial message value and the other message values input from the check node) input to one variable node.

(232) The decoding apparatus **800** may further include a memory (not illustrated) in which information on the code rate, the codeword length, and the parity-check matrix of the LDPC code is previously stored, and the LDPC decoder **810** may perform the LDPC encoding using this information. However, this is only an example, and thus, corresponding information may be provided from the transmitting side. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (a) present in every M-th column.

(233) FIG. 10 is a block diagram illustrating a configuration of a receiving apparatus according to an exemplary embodiment. As illustrated in FIG. 10, the receiving apparatus **1000** includes a demodulator **1010**, a deinterleaver **1020**, an LDPC decoder **1030** and a BCH decoder **1040**.

(234) The demodulator **1010** receives and demodulates a signal transmitted from the transmitting apparatus (for example, **700** of FIG. 7). In detail, the demodulator **1010** may demodulate the received signal to generate a value corresponding to an LDPC codeword and output a generated value to the deinterleaver **1020**.

(235) In this case, the value corresponding to the LDPC codeword may be represented by a channel value for the received signal. Here, a method for determining the channel value may be various and may be a method for determining an LLR value as one example.

(236) The deinterleaver **1020** may perform deinterleaving on output values of the demodulator **1010** and output deinterleaved output values to the LDPC decoder **1030**.

(237) In detail, the deinterleaver **1020** is a component corresponding to the interleaver **730** of the transmitting apparatus **700** and may perform an operation corresponding to the interleaver **730**. That is, the deinterleaver **1020** may inversely apply the interleaving scheme applied to the interleaver **730** to deinterleave LLR values output from the demodulator **1010**.

(238) However, in some cases, when the interleaver **730** is omitted in the transmitting apparatus **700**, the deinterleaver **1020** may be omitted.

(239) The LDPC decoder **1030** may use output values of the deinterleaver **1020** to perform the LDPC decoding and output LDPC decoded bits to the BCH decoder **1040**. Here, the LDPC decoded bits may be a BCH codeword.

(240) In detail, the LDPC decoder **1030** is a component corresponding to the LDPC encoder **720** of the transmitting apparatus **700**, and may perform the LDPC decoding based on a parity-check matrix. The

LDPC decoder **1030** of FIG. **10** may be implemented as the LDPC decoder **810** described with reference to FIG. **8**. That is, the LDPC decoder **1030** may perform the LDPC decoding using the parity-check matrix in which an information word sub-matrix is defined by above Tables 9 to 18 depending on a code rate and the parity sub-matrix has a dual diagonal structure.

(241) The BCH decoder **1040** may perform the BCH decoding on values output from the LDPC decoder **1030**.

(242) In detail, the BCH decoder **1040** is a component corresponding to the BCH encoder **710** of the transmitting apparatus **700** and may perform the BCH decoding on a BCH codeword output from the LDPC decoder **1030** to generate the bits transmitted from the transmitting apparatus **700**. However, in some cases, when the BCH encoder **710** is omitted in the transmitting apparatus **700**, the BCH decoder **1040** may be omitted.

(243) The receiving apparatus **1000** may pre-store various parameters which are used for decoding and interleaving. Here, the parameters used for decoding may be information on a code rate and a codeword length of the BCH code and information on a code rate, a codeword length and a parity-check matrix of the LDPC code. Further, the parameters used for deinterleaving may be information on a deinterleaving rule. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

(244) In this case, each component configuring the receiving apparatus **1000** may be operated using the parameters.

(245) Meanwhile, although not illustrated, in some cases, the receiving apparatus **1000** may further include a controller (not illustrated) for controlling an operation of the receiving apparatus **1000**.

(246) In this case, the controller (not illustrated) may provide the information on the code rate and the codeword length of the BCH codes to the BCH decoder **1040** and provide the information on the code rate, the codeword length, the parity-check matrix of the LDPC codes to the LDPC decoder **1030**.

Further, the controller (not illustrated) may also provide the information on the interleaving scheme to the deinterleaver **1020**. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (a) present in every M-th column.

(247) FIGS. **11** and **12** are diagrams for describing performances of LDPC codes according to an exemplary embodiment.

(248) FIG. **11** is a graph illustrating a BER performance of the LDPC codes according to an exemplary embodiment. In detail, each curve represents a BER performance to E_s/N_0 when the LDPC encoding is performed based on above Tables 9 to 18.

(249) FIG. **12** is a graph illustrating an FER performance of the LDPC codes according to an exemplary embodiment. In detail, each curve represents an FER performance to E_s/N_0 when the LDPC encoding is performed based on above Tables 9 to 18.

(250) As described above, when the LDPC encoding is performed based on the parity-check matrix defined according to the above exemplary embodiments, it may be appreciated that the BER/FER performance is improved.

(251) FIG. **13** is a flow chart for describing an encoding method according to an exemplary embodiment. In detail, FIG. **13** is a diagram for describing an encoding method of an encoding apparatus for performing low density parity check (LDPC) encoding.

(252) First, the LDPC encoding is performed on input bits based on a parity-check matrix to generate an LDPC codeword (**S1310**). In this case, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

(253) Meanwhile, the parity-check matrix may have the same form as that of the parity-check matrix **300** illustrated in FIG. **3**.

(254) In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix described above.

(255) Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word

sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

(256) In this case, in **S1310**, the LDPC encoding may be performed using parity-check matrices which are differently defined depending on a code rate.

(257) For example, the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15 and the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

(258) Meanwhile, a detailed method for performing the LDPC encoding is already described, and thus, duplicate descriptions thereof will be omitted.

(259) FIG. **14** is a flow chart for describing a decoding method according to an exemplary embodiment. In detail, FIG. **14** is a diagram for describing the decoding method of a decoding apparatus for performing low density parity check (LDPC) decoding.

(260) First, the LDPC decoding is performed on an LDPC codeword based on a parity-check matrix (**S1410**). Here, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

(261) For example, the LDPC decoding may be performed by passing LLR values corresponding to the LDPC codeword bits through an iterative decoding algorithm to generate information word bits.

(262) Here, the LLR values may be represented by channel values corresponding to the LDPC codeword bits by various methods.

(263) For example, the LLR values may be represented by values obtained by taking a log on a ratio of a probability that bits transmitted through a channel at the transmitting side are zero (0) and a probability that bits are one (1). Further, the LLR values may be bit values determined by a soft decision and may also be a representative value determined depending on a section to which the probability that the bits transmitted from the transmitting side are zero (0) or one (1) belongs.

(264) In this case, the transmitting side may use the LDPC encoder **610** as illustrated in FIG. **6** to generate and transmit the LDPC codeword.

(265) Meanwhile, the parity-check matrix may have the same form as that of the parity-check matrix **300** illustrated in FIG. **3**.

(266) In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix as described above.

(267) Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

(268) In this case, in **S1410**, the LDPC decoding may be performed using parity-check matrices which are differently defined depending on a code rate R.

(269) For example, the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15 and the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

(270) Meanwhile, the detailed method for performing the LDPC decoding is already described, and thus, duplicate descriptions thereof will be omitted.

(271) A non-transitory computer readable medium in which programs sequentially performing the encoding method and the decoding method according to the above exemplary embodiments are stored may be provided.

(272) The non-transitory computer readable medium is not a medium such as a register, a cache, and a memory which may store data for a short period of time but a medium which may semi-permanently store data and read by equipment. In detail, various applications or programs as described above may be stored and provided in the non-transitory computer readable medium such as a compact disc (CD), a

digital versatile disc (DVD), a hard disk, a Blu-ray disk, a universal serial bus (USB), a memory card, and a read-only memory (ROM).

(273) Further, in the foregoing block diagram illustrating the encoding apparatus, the decoding apparatus, the transmitting apparatus, and the receiving apparatus, a bus is not illustrated, but communication between each component in the encoding apparatus, the decoding apparatus, the transmitting apparatus, and the receiving apparatus may be made through the bus.

(274) Components, elements or units represented by a block as illustrated in FIGS. 6-10 may be embodied as the various numbers of hardware, software and/or firmware structures that execute respective functions described above, according to exemplary embodiments. For example, these components, elements or units may use a direct circuit structure, such as a memory, processing, logic, a look-up table, etc. that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. These components, elements or units may be specifically embodied by a module, a program, or a part of code, which contains one or more executable instructions for performing specified logic functions. Also, at least one of the above components, elements or units may further include a processor such as a central processing unit (CPU) that performs the respective functions, a microprocessor, or the like.

(275) As described above, according to the exemplary embodiments, the LDPC encoding and decoding performance may be improved.

(276) Hereinabove, although various exemplary embodiments of the inventive concept are illustrated and described, the inventive concept is not limited to the aforementioned exemplary embodiment and it is apparent that various modifications can be made to those skilled in the art without departing from the spirit of the inventive concept described in the appended claims and the modified embodiments are not to be individually understood from the technical spirit and prospects of the inventive concept.

Claims

1. A transmitting apparatus comprising: an encoder configured to encode input bits to generate parity bits using a matrix; an interleaver configured to interleave a codeword comprising the input bits and the parity bits; and a mapper configured to map bits of the interleaved codeword to constellation points, wherein the matrix is obtained based on parity indices of a low density parity check (LDPC) code, a code rate of the LDPC code being 6/15 and a code length of the LDPC code being 64800 bits, and wherein the parity indices are represented by a table below: TABLE-US-00019 1606 3402 4961 6751 7132 11516 12300 12482 12592 13342 13764 14123 21576 23946 24533 25376 25667 26836 31799 34173 35462 36153 36740 37085 37152 37468 37658 4621 5007 6910 8732 9757 11508 13099 15513 16335 18052 19512 21319 23663 25628 27208 31333 32219 33003 33239 33447 36200 36473 36938 37201 37283 37495 38642 16 1094 2020 3080 4194 5098 5631 6877 7889 8237 9804 10067 11017 11366 13136 13354 15379 18934 20199 24522 26172 28666 30386 32714 36390 37015 37162 700 897 1708 6017 6490 7372 7825 9546 10398 16605 18561 18745 21625 22137 23693 24340 24966 25015 26995 28586 28895 29687 33938 34520 34858 37056 38297 159 2010 2573 3617 4452 4958 5556 5832 6481 8227 9924 10836 14954 15594 16623 18065 19249 22394 22677 23408 23731 24076 24776 27007 28222 30343 38371 3118 3545 4768 4992 5227 6732 8170 9397 10522 11508 15536 20218 21921 28599 29445 29758 29968 31014 32027 33685 34378 35867 36323 36728 36870 38335 38623 1264 4254 6936 9165 9486 9950 10861 11653 13697 13961 15164 15665 18444 19470 20313 21189 24371 26431 26999 28086 28251 29261 31981 34015 35850 36129 37186 111 1307 1628 2041 2524 5358 7988 8191 10322 11905 12919 14127 15515 15711 17061 19024 21195 22902 23727 24401 24608 25111 25228 27338 35398 37794 38196 961 3035 7174 7948 13355 13607 14971 18189 18339 18665 18875 19142 20615 21136 21309 21758 23366 24745 25849 25982 27583 30006 31118 32106 36469 36583 37920 2990 3549 4273 4808 5707 6021 6509 7456 8240 10044 12262 12660 13085 14750 15680 16049 21587 23997 25803 28343 28693 34393 34860 35490 36021 37737 38296 955 4323 5145 6885 8123 9730 11840 12216 19194 20313 23056 24248 24830 25268 26617 26801 28557 29753 30745 31450 31973 32839 33025 33296 35710 37366 37509 264 605 4181 4483 5156 7238 8863 10939 11251 12964 16254 17511 20017 22395 22818 23261 23422 24064 26329 27723 28186 30434 31956 33971 34372 36764

38123 520 2562 2794 3528 3860 4402 5676 6963 8655 9018 9783 11933 16336 17193 17320 19035
20606 23579 23769 24123 24966 27866 32457 34011 34499 36620 37526 10106 10637 10906 34242
1856 15100 19378 21848 943 11191 27806 29411 4575 6359 13629 19383 4476 4953 18782 24313
5441 6381 21840 35943 9638 9763 12546 30120 9587 10626 11047 25700 4088 15298 28768 35047
2332 6363 8782 28863 4625 4933 28298 30289 3541 4918 18257 31746 1221 25233 26757 34892 8150
16677 27934 30021 8500 25016 33043 38070 7374 10207 16189 35811 611 18480 20064 38261 25416
27352 36089 38469 1667 17614 25839 32776 4118 12481 21912 37945 5573 13222 23619 31271 18271
26251 27182 30587 14690 26430 26799 34355 13688 16040 20716 34558 2740 14957 23436 32540
3491 14365 14681 36858 4796 6238 25203 27854 1731 12816 17344 26025 19182 21662 23742 27872
6502 13641 17509 34713 12246 12372 16746 27452 1589 21528 30621 34003 12328 20515 30651
31432 3415 22656 23427 36395 632 5209 25958 31085 619 3690 19648 37778 9528 13581 26965
36447 2147 26249 26968 28776 15698 18209 30683 1132 19888 34111 4608 25513 38874 475 1729
34100 7348 32277 38587 182 16473 33082 3865 9678 21265 4447 20151 27618 6335 14371 38711 704
9695 28858 4856 9757 30546 1993 19361 30732 756 28000 29138 3821 24076 31813 4611 12326
32291 7628 21515 34995 1246 13294 30068 6466 33233 35865 14484 23274 38150 21269 36411 37450
23129 26195 37653.

2. The transmitting apparatus of claim 1, wherein the matrix has a cyclic structure.

3. The transmitting apparatus of claim 1, wherein a predetermined number which is differently set according to the code rate and the code length is 108 when the code rate is 6/15 and the code length is 64800 bits.

4. The transmitting apparatus of claim 1, wherein the parity bits are generated by initiating parity bits, accumulating the input bits to the initiated parity bits based on the matrix, and performing a bit operation based on an accumulated parity bit of a parity index x and an accumulated parity bit of a parity index $x-1$ to output a parity bit of the parity index x , where x is a value among values which are greater than 0 and are less than a number of parity bits.

5. The transmitting apparatus of claim 4, wherein the matrix is represented by $q(i,j,l)$ generated based on a following relationship:

$q(i,j,l)=q(i,j,0)+Q.\text{sub.ldpc} \times 1(\text{mod } M.\text{sub.inner})$ for $0 < l < 360$ where $Q.\text{sub.ldpc}$ is a coding parameter determined by the code rate of the LDPC code and the code length of the LDPC code, $M.\text{sub.inner}$ is the number of parity bits, $q(i,j,0)$ is a parity index represented by a j -th entry in an i -th row of the table, i indicates an index of a row in the table, and j indicates an index of an entry in the row.

6. The transmitting apparatus of claim 5, wherein the input bits are accumulated to the initiated parity bits based on a following relationship,

$p.\text{sub.q}(i,w(i),l)=P.\text{sub.q}(i,w(i),1).\text{sup.}+s.\text{sub.k}$, where $s.\text{sub.k}$ indicates k -th input bit from among the input bits, $p.\text{sub.q}(i,w(i),l)$ indicates $q(i,w(i),l)$ -th parity bit from among the initiated parity bits, k is a value among values which are greater than or equal to 0 and are less than a number of the input bits, $i=\lfloor k/360 \rfloor$, $l=k \text{ mod } 360$, and $w(i)=j$.

7. A receiving apparatus comprising: a demodulator configured to demodulate a signal received from a transmitting apparatus to generate values; a deinterleaver configured to deinterleave the values; and a decoder configured to decode the deinterleaved values based on a matrix of a low density parity check (LDPC) code, a code rate of the LDPC code being 6/15 and a code length of the LDPC code being 64800 bits, wherein the matrix is obtained based on parity indices, and wherein the parity indices are represented by a table below:

TABLE-US-00020 1606 3402 4961 6751 7132 11516 12300 12482 12592
13342 13764 14123 21576 23946 24533 25376 25667 26836 31799 34173 35462 36153 36740 37085
37152 37468 37658 4621 5007 6910 8732 9757 11508 13099 15513 16335 18052 19512 21319 23663
25628 27208 31333 32219 33003 33239 33447 36200 36473 36938 37201 37283 37495 38642 16 1094
2020 3080 4194 5098 5631 6877 7889 8237 9804 10067 11017 11366 13136 13354 15379 18934 20199
24522 26172 28666 30386 32714 36390 37015 37162 700 897 1708 6017 6490 7372 7825 9546 10398
16605 18561 18745 21625 22137 23693 24340 24966 25015 26995 28586 28895 29687 33938 34520
34858 37056 38297 159 2010 2573 3617 4452 4958 5556 5832 6481 8227 9924 10836 14954 15594
16623 18065 19249 22394 22677 23408 23731 24076 24776 27007 28222 30343 38371 3118 3545 4768
4992 5227 6732 8170 9397 10522 11508 15536 20218 21921 28599 29445 29758 29968 31014 32027
33685 34378 35867 36323 36728 36870 38335 38623 1264 4254 6936 9165 9486 9950 10861 11653

13697 13961 15164 15665 18444 19470 20313 21189 24371 26431 26999 28086 28251 29261 31981
34015 35850 36129 37186 111 1307 1628 2041 2524 5358 7988 8191 10322 11905 12919 14127 15515
15711 17061 19024 21195 22902 23727 24401 24608 25111 25228 27338 35398 37794 38196 961 3035
7174 7948 13355 13607 14971 18189 18339 18665 18875 19142 20615 21136 21309 21758 23366
24745 25849 25982 27583 30006 31118 32106 36469 36583 37920 2990 3549 4273 4808 5707 6021
6509 7456 8240 10044 12262 12660 13085 14750 15680 16049 21587 23997 25803 28343 28693
34393 34860 35490 36021 37737 38296 955 4323 5145 6885 8123 9730 11840 12216 19194 20313
23056 24248 24830 25268 26617 26801 28557 29753 30745 31450 31973 32839 33025 33296 35710
37366 37509 264 605 4181 4483 5156 7238 8863 10939 11251 12964 16254 17511 20017 22395 22818
23261 23422 24064 26329 27723 28186 30434 31956 33971 34372 36764 38123 520 2562 2794 3528
3860 4402 5676 6963 8655 9018 9783 11933 16336 17193 17320 19035 20606 23579 23769 24123
24966 27866 32457 34011 34499 36620 37526 10106 10637 10906 34242 1856 15100 19378 21848 943
11191 27806 29411 4575 6359 13629 19383 4476 4953 18782 24313 5441 6381 21840 35943 9638
9763 12546 30120 9587 10626 11047 25700 4088 15298 28768 35047 2332 6363 8782 28863 4625
4933 28298 30289 3541 4918 18257 31746 1221 25233 26757 34892 8150 16677 27934 30021 8500
25016 33043 38070 7374 10207 16189 35811 611 18480 20064 38261 25416 27352 36089 38469 1667
17614 25839 32776 4118 12481 21912 37945 5573 13222 23619 31271 18271 26251 27182 30587
14690 26430 26799 34355 13688 16040 20716 34558 2740 14957 23436 32540 3491 14365 14681
36858 4796 6238 25203 27854 1731 12816 17344 26025 19182 21662 23742 27872 6502 13641 17509
34713 12246 12372 16746 27452 1589 21528 30621 34003 12328 20515 30651 31432 3415 22656
23427 36395 632 5209 25958 31085 619 3690 19648 37778 9528 13581 26965 36447 2147 26249
26968 28776 15698 18209 30683 1132 19888 34111 4608 25513 38874 475 1729 34100 7348 32277
38587 182 16473 33082 3865 9678 21265 4447 20151 27618 6335 14371 38711 704 9695 28858 4856
9757 30546 1993 19361 30732 756 28000 29138 3821 24076 31813 4611 12326 32291 7628 21515
34995 1246 13294 30068 6466 33233 35865 14484 23274 38150 21269 36411 37450 23129 26195
37653.

8. The receiving apparatus of claim 7, wherein the matrix has a cyclic structure.
